

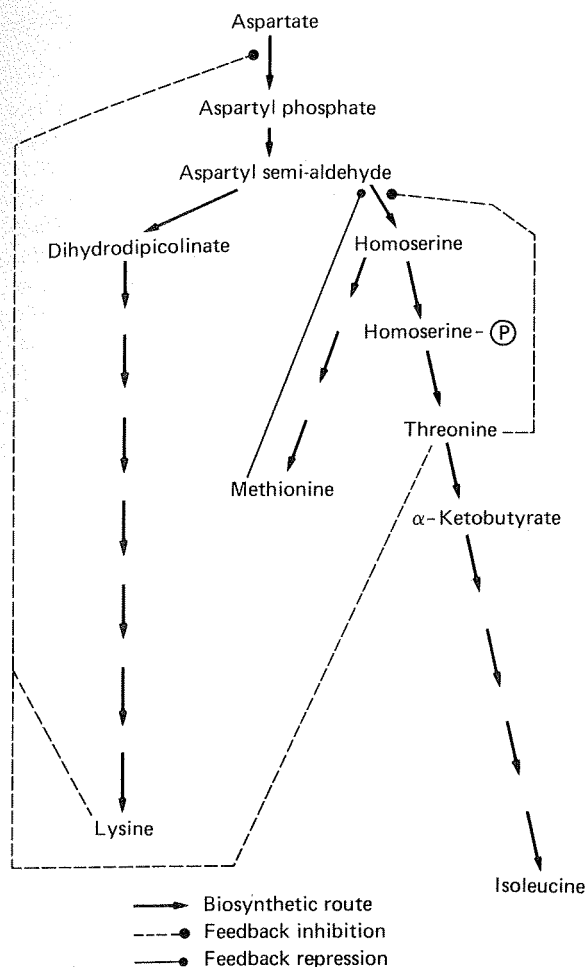


FIG. 3.14. The control of the aspartate family of amino acids in *C. glutamicum*.

first enzyme in the pathway, is controlled by the concerted feedback inhibition of lysine and threonine. Homoserine dehydrogenase is subject to feedback inhibition by threonine and repression by methionine. The first enzyme in the route from aspartate semialdehyde to lysine is not subject to feedback control. Thus, the control system found in *C. glutamicum* is a relatively simple one. Nakayama *et al.* (1961) selected a homoserine auxotroph of *C. glutamicum*, by the penicillin selection and replica plating method, which produced lysine in a medium containing a low level of homoserine, or threonine plus methionine. The mutant lacked homoserine dehydrogenase which allowed aspartic semi-aldehyde to be converted solely to lysine and the resulting lack of threonine removed the

concerted feedback inhibition of aspartokinase. Kinoshita and Nakayama (1978) quoted the homoserine auxotroph, *C. glutamicum* 901, as producing 44 g dm^{-3} lysine.

The control of the production of arginine in *C. glutamicum* is shown in Fig. 3.15. The major control of the pathway is the feedback inhibition of the second enzyme in the sequence, acetylglutamic acid phosphorylating enzyme, although the first enzyme may also be subject to regulation.

Kinoshita *et al.* (1957b) isolated a citrulline requiring auxotroph of *C. glutamicum* which would accumulate ornithine at a molar yield of 36% from glucose, in the presence of limiting arginine and excess biotin. The mutant lacked the enzyme converting ornithine to citrulline which resulted in the cessation of arginine synthesis and, therefore, the removal of the control of the pathway.

Inosine monophosphate has been shown to demonstrate flavour-enhancing qualities and is produced commercially by the chemical phosphorylation of ino-

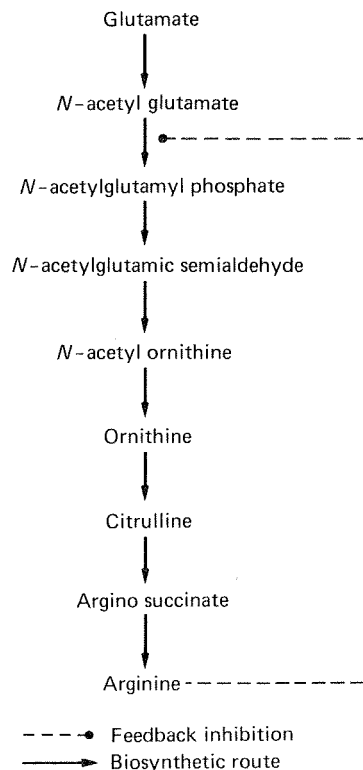


FIG. 3.15. The control of the biosynthesis of arginine in *C. glutamicum*.

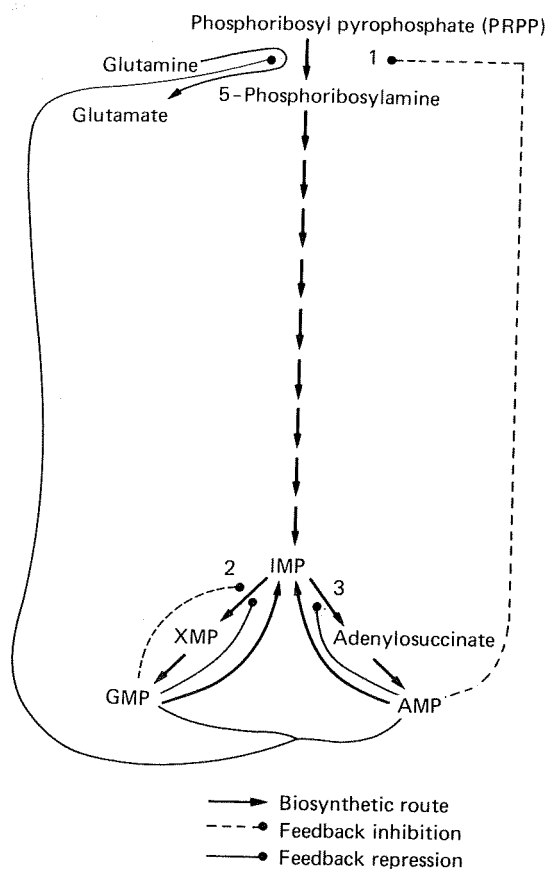


FIG. 3.16. Control of the biosynthesis of the purine nucleotides in *Bacillus subtilis*. 1. Reaction catalysed by PRPP amidinotransferase. 2. Reaction catalysed by IMP dehydrogenase. 3. Reaction catalysed by adenylosuccinate synthase. AMP = Adenosine monophosphate. IMP = Inosine monophosphate. XMP = Xanthine monophosphate. GMP = Guanosine monophosphate.

sine (Shibai *et al.* 1978) which is produced from auxotrophic strains of *Bacillus subtilis*. The control of the production of purine nucleotides is shown in Fig. 3.16. The main sites of control shown in Fig. 3.16 are:

- Phosphoribosyl pyrophosphate (PRPP) amidinotransferase (the first enzyme in the sequence) is feedback inhibited by AMP but only very slightly by GMP.
- The synthesis of PRPP amidinotransferase is repressed by the co-operative action of AMP and GMP, as are the syntheses of the other enzymes indicated in Fig. 3.16.
- IMP dehydrogenase is feedback inhibited and repressed by GMP.

- Adenylosuccinate synthase is repressed by AMP but is not significantly inhibited.

Mutants which are auxotrophic for AMP or doubly auxotrophic for AMP and GMP have been isolated which will excrete inosine at levels of up to 15 g dm^{-3} (Demain, 1978). AMP auxotrophs, lacking adenylosuccinate synthase activity, require the addition of small quantities of adenosine but will accumulate inosine due to the removal of the inhibition and co-operative repression of PRPP amidinotransferase, as shown in Fig. 3.17. AMP and GMP double auxotrophs will produce inosine due to the removal of the controls normally imposed by the two end products, as illustrated in Fig. 3.18. Such double auxotrophs require the addition of both adenosine and guanosine, in small concentrations.

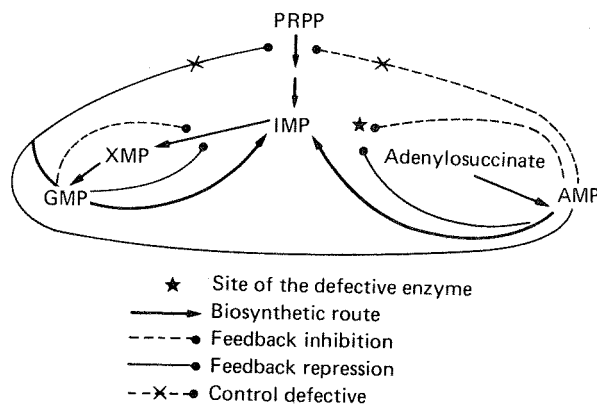


FIG. 3.17. The control of the synthesis of purine nucleotides in a mutant with a defective adenylosuccinate synthase.

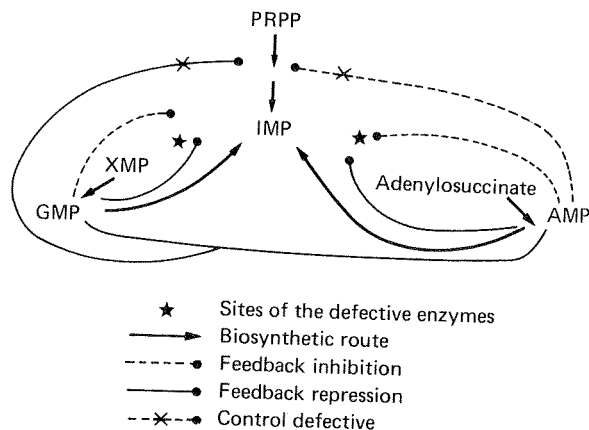


FIG. 3.18. The control of the synthesis of purine nucleotides in a mutant deficient in adenylosuccinate synthase and IMP dehydrogenase activities.

THE ISOLATION OF MUTANTS THAT DO NOT RECOGNIZE THE PRESENCE OF INHIBITORS AND REPRESSORS

The use of auxotrophic mutants has resulted in the production of many microbial products in large concentrations, but, obviously, such mutants are not suitable for the synthesis of products which control their own synthesis independently. A hypothetical example is shown in Fig. 3.19 where the end product P controls its own biosynthesis by feedback inhibition of the first enzyme in the pathway. If it is required to produce the intermediate F in large concentrations then this may be achieved by the isolation of a mutant auxotrophic for P, blocked between F and P. However, if P is required to be synthesized in large concentrations it is quite useless to produce an auxotrophic mutant. The solution to this problem is to modify the organism such that the first enzyme in the pathway no longer recognizes the presence of inhibiting levels of P. The isolation of mutants altered in the recognition of control factors has been achieved principally by the use of two techniques:

- (i) The isolation of analogue resistant mutants.
- (ii) The isolation of revertants.

An analogue is a compound which is very similar in structure to another compound. Analogues of amino acids and nucleotides are frequently growth inhibitory, and their inhibitory properties may be due to a number of possible mechanisms. For example, the analogue may be used in the biosynthesis of macromolecules resulting in the production of defective cellular components. In some circumstances the analogue is not incorporated in place of the natural product but interferes with its biosynthesis by mimicking its control properties. For example, consider the pathway illustrated in Fig. 3.19 where the end product, P, feedback inhibits the first enzyme in the pathway. If P^* were an analogue of P (which could not substitute for P in biosynthesis) and were to inhibit the first enzyme in a similar way to P, then the biosynthesis of P may be prevented by P^* which could result in the inhibition of the growth of the organism.

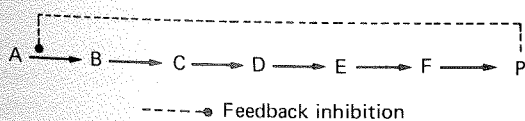


Fig. 3.19. The control of the production of an end product P.

Mutants may be isolated which are resistant to the inhibitory effects of the analogue and, if the site of toxicity of the analogue is the mimicking of the control properties of the natural product, such mutants may overproduce the compound to which the analogue is analogous. To return to the example of the biosynthesis of P where P^* is inhibitory due to its mimicking the control properties of P; a mutant may be isolated which may be capable of growing in the presence of P^* due to the fact that the first enzyme in the pathway is no longer susceptible to inhibition by the analogue. The modified enzyme of the resistant mutant may not only be resistant to inhibition by the analogue but may also be resistant to the control effects of the natural end product, P, resulting in the uninhibited production of P. If the control system were the repression of enzyme synthesis, then the resistant mutant may be modified such that the enzyme synthesis machinery does not recognize the presence of the analogue. However, the site of resistance of the mutant may not be due to a modification of the control system; for example, the mutant may be capable of degrading the analogue, in which case the mutant would not be expected to overproduce the end product. Thus, analogue resistant mutants may be expected to overproduce the end product to which the analogue is analogous provided that:

- (i) The toxicity of the analogue is due to its mimicking the control properties of the natural product.
- (ii) The site of resistance of the resistant mutant is the site of control by the end product.

Resistant mutants may be isolated by exposing the survivors of a mutation treatment to a suitable concentration of the analogue in growth medium and purifying any colonies which develop. Sermonti (1969) described a method to determine the suitable concentration. The organism was exposed to a range of concentrations of the toxic analogue by inoculating each of a number of agar plates containing increasing levels of the analogue with 10^6 to 10^9 cells. The plates were incubated for several days and examined to determine the lowest concentration of analogue which allowed only a very few isolated colonies to grow, or completely inhibited growth. The survivors of a mutation treatment may then be challenged with the pre-determined concentration of the analogue on solid medium. Colonies which develop in the presence of the analogue may be resistant mutants.

Szybalski (1952) constructed a method of exposing

the survivors of a mutation to a range of analogue concentrations on a single plate. Known as the gradient plate technique, it consists of pouring 20 cm³ of molten agar medium, containing the analogue, into a slightly slanted petri dish and allowing the agar to set at an angle. After the agar has set, a layer of medium not containing the analogue is added and allowed to set with the plate level. The analogue will diffuse into the upper layer giving a concentration gradient across the plate and the survivors of a mutation treatment may be spread over the surface of the plate and incubated. Resistant mutants should be detected as isolated colonies appearing beyond a zone of confluent growth, as indicated in Fig. 3.20. Whichever method is used for the isolation of analogue-resistant mutants, great care should be taken to ensure that the isolates are genuinely resistant to the analogue by streaking them, together with analogue-sensitive controls, on both analogue-supplemented and analogue-free media. The resistant isolates should then be screened for the production of the desired compound by overlaying them with a bacterial strain requiring the compound; producers may then be recognized by a halo of growth of the indicator strain.

Sano and Shiio (1970) investigated the use of lysine analogue-resistant mutants of *Brevibacterium flavum* for the production of lysine. The control of the biosynthesis of the aspartate family of amino acids in *B. flavum* is as illustrated for *C. glutamicum* in Fig. 3.14. The main control of lysine synthesis is the concerted feedback inhibition of aspartokinase by lysine and threonine. Sano and Shiio demonstrated that *S*-(2-aminoethyl) cysteine (AEC) completely inhibited the growth of *B. flavum* in the presence of threonine, but only partially in its absence. Also, the inhibition by AEC and threonine could be reversed by the addition of lysine. This evidence suggested that the inhibitory effect of AEC was due to its mimicking lysine in the concerted inhibition of aspartokinase. AEC-resistant mutants were isolated by plating the survivors of a mutation treatment on minimal agar containing 1 mg cm⁻³ of both AEC and threonine. A relatively large number of the resistant isolates accumulated lysine, the best producers synthesizing more than 30 g dm⁻³. Investigation of the lysine producers indicated that their aspartokinases had been desensitized to the concerted inhibition by lysine and threonine.

The development of an arginine-producing strain of *B. flavum* by Kubota *et al.* (1973) provides an excellent example of the selection of a series of mutants resistant to increasing levels of an analogue. The control of the

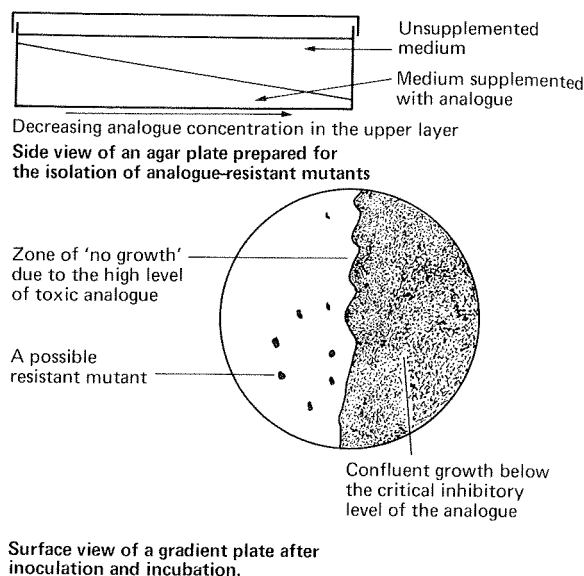


Fig. 3.20. The gradient plate technique for the isolation of analogue-resistant mutants.

biosynthesis of arginine in *B. flavum* is similar to that shown for *C. glutamicum* in Fig. 3.15. Kubota *et al.* selected mutants resistant to the arginine analogue, 2-thiazolealanine, and the genealogy of the mutants is shown in Fig. 3.21. Strain number 352 produced 25.3 g dm⁻³ arginine. Presumably, the mutants were altered in the susceptibility of the second enzyme in the pathway to inhibition by arginine.

A classic example of the rationale of analogue resistant mutant isolation is seen in the development of biotin overproducing strains. Currently, biotin is produced commercially by chemical synthesis, but considerable effort has been (and is being) expended to develop a competitive biotechnological process. The screening of natural isolates for their ability to accumulate biotin-vitamins led to the isolation of *Bacillus sphaericus* (Ogata *et al.*, 1965). Repression by biotin was shown to be an important control mechanism and, thus, attempts were made to isolate mutants resistant to biotin analogues. Mutants of *B. sphaericus* resistant to acidomycin (ACM) and/or 5-(2-thienyl)-*n*-valeric acid (TVA) were capable of synthesizing up to 11 times (0.4 mg cm⁻³) the biotin level of the wild-type (Tanaka *et al.*, 1988). More recently, Sakurai *et al.* (1993) isolated dual ACM/TVA resistant strains of *Serratia marcescens* capable of synthesizing 20 mg cm⁻³ biotin. The use of genetic engineering techniques to further develop these

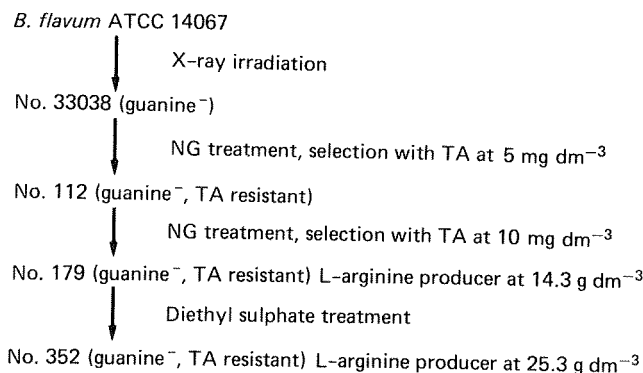


FIG. 3.21. The genealogy of L-arginine-producing mutants of *B. flavum*. TA, thiazolealanine; NG, *N*-methyl-*N'*-nitro-*N*-nitroso-guanidine (Kubota *et al.*, 1973).

strains is considered in a later section of this chapter.

The second technique used for the isolation of mutants altered in the recognition of control factors is the isolation of revertant mutants. Auxotrophic mutants may revert to the phenotype of the mutant 'parent'. Consider the hypothetical pathway illustrated in Fig. 3.19 where P controls its own production by feedback inhibiting the first enzyme (*a*) of the pathway. A mutant does not produce the enzyme, *a*, and is, therefore, auxotrophic for P. However, a revertant of the mutant produces large concentrations of P. The explanation of the behaviour of the revertant is that, with two mutations having occurred at loci concerned with the production of enzyme *a*, the enzyme of the revertant is different from the enzyme of the original prototrophic strain and is not susceptible to the control by P. Revertants may occur spontaneously or mutagenic agents may be used to increase the frequency of occurrence, but the recognition of the revertants would be achieved by plating millions of cells on medium which would allow the growth of only the revertants, i.e. in the above example, on medium lacking P.

Shiio and Sano (1969) investigated the use of prototrophic revertants of *B. flavum* for the production of lysine. These workers isolated prototrophic revertants from a homoserine dehydrogenase-defective mutant. The revertants were obtained as small-colony forming strains and produced up to 23 g dm⁻³ lysine. The overproduction of lysine was shown to be due to the very low level of homoserine dehydrogenase in the revertants which, presumably, resulted in the synthesis of threonine and methionine in quantities sufficient for some growth, but insufficient to cause inhibition or repression.

Mutant isolation programmes for the improvement

of strains producing primary metabolites have not relied on the use of only one selection technique. Most projects employed a number of methods including the selection of natural variants and the selection of induced mutants by a variety of means. The selection of bacteria overproducing threonine provides a good example of the use of a variety of selection techniques. Attempts to isolate auxotrophic mutants of *C. glutamicum* producing threonine were unsuccessful despite the fact that productive auxotrophic strains of *Escherichia coli* had been isolated. Huang (1961) demonstrated threonine production at a level of 2–4 g dm⁻³ by a diaminopimelate and methionine double auxotroph of *E. coli*. Kase *et al.* (1971) isolated a triple auxotrophic mutant of *E. coli* which required diaminopimelate, methionine and isoleucine and produced between 15 and 20 g dm⁻³ threonine. The control of the production of the aspartate family of amino acids in *E. coli* is shown in Fig. 3.22 and that in *C. glutamicum* in Fig. 3.14. The mechanism of control in *E. coli* involves a system of isoenzymes, three isoenzymic forms of aspartokinase and two of homoserine dehydrogenase, under the influence of different end products. However, in *C. glutamicum* control is effected by the concerted inhibition of a single aspartokinase by threonine and lysine; by the inhibition of homoserine dehydrogenase by threonine and the repression of homoserine dehydrogenase by methionine. Thus, the control of homoserine dehydrogenase may not be removed by auxotrophy without the loss of threonine production. However, in *E. coli* methionine auxotrophy would remove control of the methionine-sensitive homoserine dehydrogenase and aspartokinase which would allow threonine production, despite the control of the threonine-sensitive isoenzymes by threonine. *E. coli* mutants

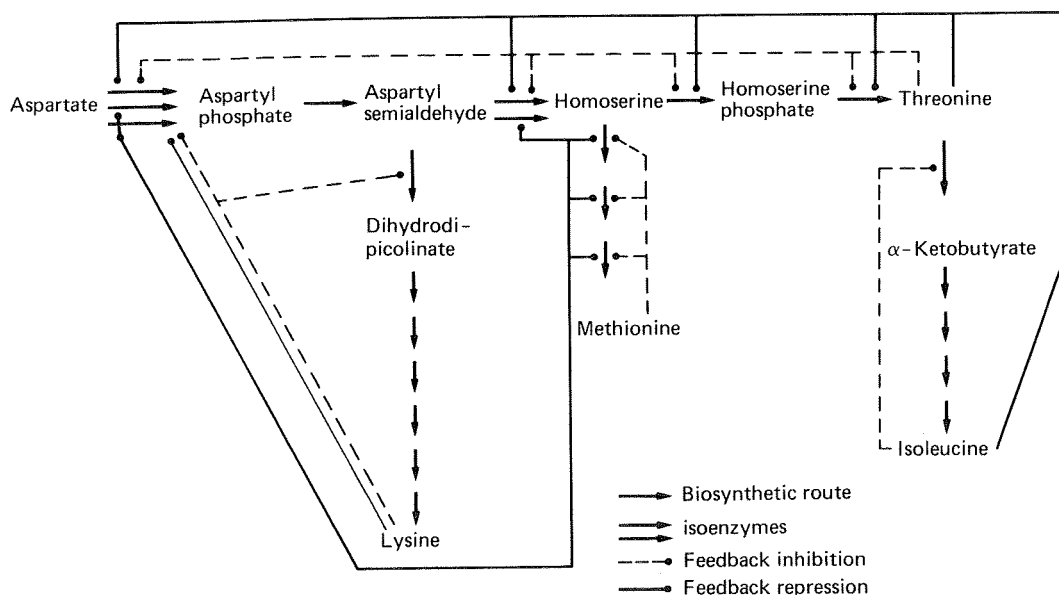


FIG. 3.22. Control of the aspartate family of amino acids in *Escherichia coli*.

also lacking lysine and isoleucine would be relieved of the control of the lysine-sensitive aspartokinase and the degradation of threonine to isoleucine.

The production of threonine by *C. glutamicum* has been achieved by the use of combined auxotrophic and analogue resistant mutants. A good example of the approach is given by Kase and Nakayama (1972) who obtained stepwise improvements in productivity by the imposition of resistance to α -amino- β -hydroxy valeric acid (a threonine analogue) and *S*-(β -aminoethyl)-L-

cysteine (a lysine analogue) on a methionine auxotroph of *C. glutamicum*. The genealogy of the mutants is shown in Fig. 3.23. The analogue-resistant strains were shown to be altered in the susceptibility of aspartokinase and homoserine dehydrogenase to control, and the lack of methionine removed the repression control of homoserine dehydrogenase. The use of recombinant DNA technology has resulted in the construction of far more effective threonine producers and these strains are considered in a later section of this chapter.

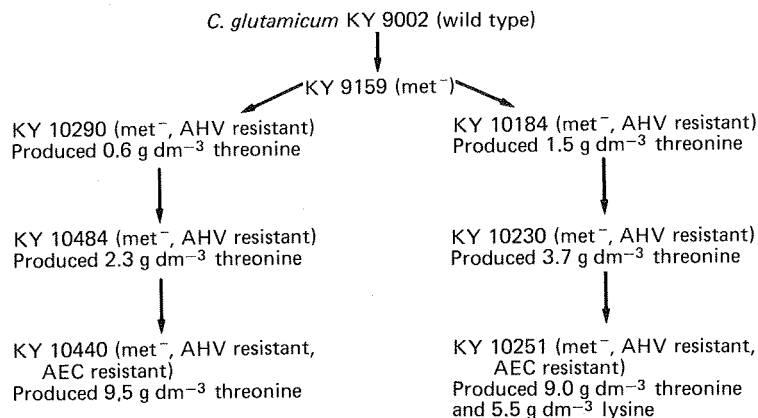


FIG. 3.23. The genealogy of mutants of *C. glutamicum* producing L-threonine or L-threonine plus L-lysine. AHV, α -amino- β -hydroxy valeric acid; AEC, *S*-(β -aminoethyl)-L-cysteine (Kase and Nakayama, 1972).

The development of strains producing guanosine in high levels provides a further example of the use of auxotrophs and analogue-resistant mutants in the same programme. Guanosine production has been achieved using auxotrophic and analogue resistant mutants of *B. subtilis*. The control of the production of purine nucleotides in *B. subtilis* is shown in Fig. 3.16 (Demain, 1978). The major points of control to be modified to achieve guanosine overproduction are:

- (i) Removal of AMP inhibition of phosphoribosyl pyrophosphate amidino transferase by adenine auxotrophy.
- (ii) Elimination of GMP reductase.
- (iii) Bypassing the inhibition of IMP dehydrogenase by GMP.

This approach is demonstrated in the work of Shiio (cited by Demain, 1978) who obtained analogue-resistant mutants from an adenine auxotroph of *B. subtilis*. Mutants were isolated in two stages — first, for resistance to low levels of 8-azaguanine (an analogue of GMP) and then for resistance to high levels of the analogue. The resulting mutant produced 9 g dm^{-3} guanosine due to the removal of control of phosphoribosyl pyrophosphate amidinotransferase and IMP dehydrogenase.

**The isolation of induced mutants
producing improved yields of secondary
metabolites where directed selection
is difficult to apply**

The discussion so far has considered the isolation of mutants producing products whose biosynthesis and control have been sufficiently understood to prepare 'blueprints' of the desirable mutants which have enabled the construction of suitable selection procedures. In contrast, important secondary metabolites were being produced long before their biosynthetic pathways, and certainly the control of those pathways, had been elucidated. Thus, strain improvement programmes had to be developed without this fundamental knowledge which meant that they depended on the random selection of the survivors of mutagen exposure. Elander and Vournakis (1987) described these techniques as "hit or miss methods that require brute force, persistence and skill in the art of microbiology". However, despite the limited knowledge underlying these approaches they were extremely effective in increasing the yields of antibiotics, as illustrated in Table 3.5 (Riviere, 1977).

TABLE 3.5. *Improvement of antibiotic yields during the first 20 years of antibiotic development (Riviere, 1977)*

Antibiotic	Initial yield at time of discovery (units cm^{-3})	Improved yield in France, 1972 (units cm^{-3})
Penicillin	20 (1943)	12,000–15,000
Streptomycin	50 (1945)	12,000–15,000
Chlortetracycline	200 (1948)	12,000–15,000
Erythromycin	100 (1955)	3000

More rational approaches have been developed which reduce the empirical nature of strain-improvement programmes. These developments include the streamlining of the empirical techniques and the use of more directed selection methods. Before discussing the attempts at directed selection the empirical approach will be considered along with the attempts made to improve this approach, including miniaturized programmes.

The empirical approach to strain improvement involves subjecting a population of the micro-organism to a mutation treatment and then screening a proportion of the survivors of the treatment for improved productivity. The assessment of the chosen survivors was usually carried out in shake flasks, resulting in the procedure being costly, both in terms of time and personnel. According to Fantini (1966) the two questions which arise in the design of such programmes are:

- (i) How many colonies from the survivors of a mutation treatment should be isolated for testing?
- (ii) Which colonies should be isolated?

In attempting to answer the first question Fantini dismissed statistical approaches as impractical and claimed that the number of colonies isolated for testing is determined by the practical limiting factors of personnel, incubator and shaker space and time. However, Davies (1964) demonstrated that a statistical approach could give valuable guidelines for the efficient utilization of physical resources in strain-development programmes. Davies based a computer simulation of a mutation and screening programme on the availability of 200 shaker spaces and practical results of error variance and the distribution of yield likely to occur amongst the mutants. He assumed that the majority of the progeny of a mutation would give a small, rather than a large, increase in productivity and it would, therefore, be more feasible to screen a small number in the hope of obtaining a small increase rather than

screen a large number in the hope of obtaining a large increase. The major difficulty inherent in this approach is the error involved in determining small increases which implies the replication of screening tests and, therefore, the use of more facilities.

Davies used the computer simulation to investigate the merits of replication in screening programmes and finally proposed the use of a two-stage scheme where mutants were tested singly in the first stage and then the better producers were tested in quadruplicate in the second stage. Davies concluded that such two-stage schemes were adequate over a wide range of conditions, although a one-stage screen could be used if the testing error were small and the frequency of occur-

rence of favourable types high, and a three-stage screen if the testing error were high and the frequency of occurrence of favourable types low. A screening programme based on Davies' proposals is shown in Fig. 3.24.

The answer to Fantini's second question (which colonies should be isolated?) is extremely difficult in the field of secondary metabolites and in Davies' scheme the colonies were chosen at random. The selection of colonies on the basis of changed morphology has been considered by a number of workers and it appears that this is an undesirable technique. Elander (1966) demonstrated that it was preferable to isolate normal morphological types as, although a morphologi-

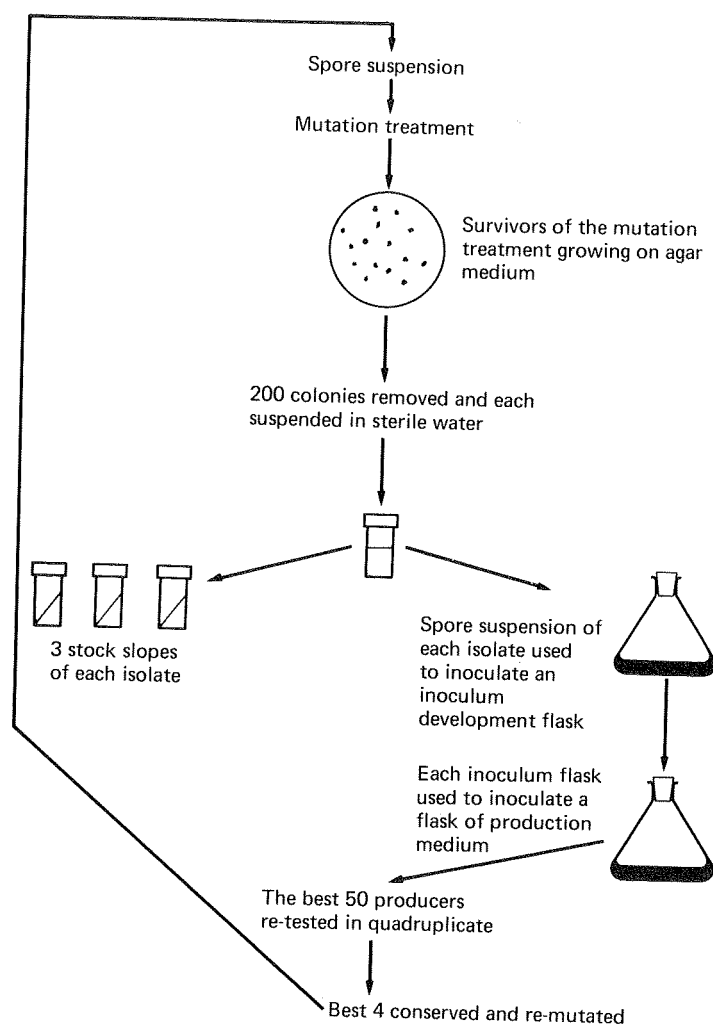


FIG. 3.24. A strain-improvement programme for a secondary metabolite producing culture (Davies, 1964).

cal variant may be a superior producer, it might require considerable fermentation development to materialize the increased production. Also, Alikhanian (1962) claimed that the vast majority of morphological mutants of the antibiotic producing actinomycetes tested were inferior producers. The most common type of shake flask programme quoted is similar to that of Davies where the choice of colonies is random. However, there are now many reports of screens which miniaturize the procedure for the improvement of secondary metabolite producing strains. Such miniaturized systems are designed to enable the productivity of all (or a significant proportion) of the survivors to be assessed which should eliminate (or reduce) the problem of choice of colonies for assessment and increase the throughput of cultures.

The basis of the miniaturized techniques is to grow the survivors of the mutation treatment either in a very low volume of liquid medium or on solidified (agar) medium. If the product is an antibiotic, the agar-grown colonies may be overlayed with an indicator organism sensitive to the antibiotic produced, allowing assay to be done *in situ*. The level of antibiotic is assessed by the degree of inhibition of the overlayed indicator. The system is simple to apply to strains producing low levels of antibiotic but must be modified to allow the screening of high producers where very large zones of inhibition would be obtained. Also a system should be used to free the superior producers from contaminating indicator organisms. Dulaney and Dulaney (1967) used overlay techniques in the isolation of mutants producing chlortetracycline. Mutated spores were cultured on an agar medium in petri dishes for 6 days and then covered with pieces of sterile cellophane. An overlay of agar containing the indicator organism was then added and the plates incubated overnight. The mutant colonies were kept free from contamination by the cellophane and the size of the inhibition zone could be controlled by the depth of the base layer, the age of the colonies when over-layed, the depth of the overlay and the temperature of incubation. The system was calibrated initially such that a single colony of a non-mutated strain would not produce an inhibition zone but that two such adjacent colonies would. In practice, the size of the inhibition zone was controlled by the depth of the overlay. Dulaney and Dulaney obtained a far greater enrichment in the number of desired phenotypes by the overlay technique than by random selection and testing in liquid medium.

Ichikawa *et al.* (1971) screened for the increased production of the antibiotic kasugamycin by a *Streptomyces* sp., using a miniaturized technique termed the

agar piece method. In order to prevent interference between colonies, mutated spores were grown on plugs of agar which were then placed on assay plates containing agar seeded with the indicator organism, levels of the antibiotic being determined by the size of inhibition zones. By combining this technique with a medium improvement programme the authors obtained a ten-fold increase in productivity. Ichikawa's method was modified by Ditchburn *et al.* (1974) for the isolation of *Aspergillus nidulans* mutants synthesizing improved levels of penicillin. These workers obtained promising results using the technique and claimed that its potential for the recovery of higher-yielding mutants, for a given expenditure of labour time, was greater than the shake-flask method. The level of production of penicillin by *A. nidulans* is very small compared with *P. chrysogenum* and the technique would have to be modified considerably for commercial use.

Ball and McConagle (1978) developed a miniaturized technique suitable for the assay of penicillin production by 'high-yielding' strains of *P. chrysogenum* (producing up to 6000 units cm^{-3}). These workers highlighted the design of the solidified medium as a critical factor in the optimization of the method. They claimed that the growth of a colony on agar-solidified medium would be unlikely to modify the medium to the same extent as the growth of the organism on the same medium in submerged liquid culture. Thus, the nutrient-limiting conditions that favour the onset of antibiotic production might not be achieved by the growth of the culture on solid medium which would not allow the full production potential of the culture to be detected. However, nutrient-limiting conditions were achieved in the solidified medium by omitting the main carbon source and reducing the corn steep-liquor content. Mutated spores were plated over the surface of petri dishes of the nutrient limited medium such that ten to twenty colonies developed per plate. The time of incubation of the plates was not quoted but "when the colonies were of a size suitable for accurate measurement" a pasteurized spore suspension of *B. subtilis* containing 0.16 units cm^{-3} of penicillinase (to limit the size of the inhibition zones) was dispersed over the surface of the dish. The plates were then incubated for 18 to 24 hours and the inhibition zones examined. Suitable colonies were freed of contaminating *B. subtilis* by culturing on nutrient agar containing sodium penicillin and streptomycin sulphate. The use of this technique enabled three operators to scan 15,000 survivors from ultraviolet irradiated spore populations in 3 months.

The major disadvantages of the miniaturized solidi-

fied medium technique approach is that productivity expressed on the solid medium may not be expressed in subsequent liquid culture and conversely, colonies not showing activity on solid media may be highly productive in liquid medium. Despite these limitations the above workers have demonstrated that the approach has considerable merit and Ball (1978) claimed that the increase in throughput may be as much as 20 times that of a conventional shake-flask programme. Some recent work by Bushell's group (Pickup *et al.*, 1993) provides an interesting insight into the problem of certain streptomycete isolates producing a secondary metabolite on agar but not in liquid culture. It is important to appreciate that the strains were natural isolates and not the survivors of mutation treatments but the conclusions do have relevance to mutant development. Working on the premise that non-production in liquid medium may be due to a more fragmented morphology these strains were subjected to a filtration enrichment procedure. Liquid cultures were filtered through a linen filter and the filtrate and retentate mycelium re-cultured in fresh medium. The procedure was then repeated sequentially. The enriched retentate mycelium of several isolates which were previously unable to produce antibiotic in liquid culture gave rise to stable filamentous types synthesizing in liquid medium. Although this approach could not be used routinely in a high throughput mutant screen it may be applied to a few high-producing strains which have not fulfilled their promise (detected in an agar screen) in liquid media.

An excellent example of a miniaturized screening programme is given by the work of Dunn-Coleman *et al.* (1991) of Genencor International. These workers developed a low-volume liquid medium system to isolate mutants of *Aspergillus niger* var. *awamori* capable of improved secretion of bovine chymosin. The *Aspergillus* strain had been constructed using recombinant DNA technology and this construction will be considered in a later section of this Chapter. Mutated spore suspensions were diluted and inoculated into 96-well microtitre plates using robots. The dilutions were such that each well should have contained one viable spore. The plates were incubated in a static incubator and then harvested using a robotic pipetting station and assayed for product. Typically, 50–60,000 mutated viable spores were assessed in each screen. The most promising 10–50 strains from the miniaturized screen were then tested in shake flask culture. The best producer from the shake-flask screen was then used as the starter for a further round of mutagenesis. The results of seven rounds of mutation and selection are shown in Table 3.6. Investigation of the improved strains showed

TABLE 3.6. Chymosin production by NTG mutated strains of *Aspergillus niger* var *awamori* (Dunn-Coleman *et al.*, 1991)

Rounds of NTG mutagenesis and recurrent selection	Chymosin concentration ($\mu\text{g cm}^{-3}$)	
	Microtitre plate	Shake flask
Parent	42	286
First	50	273
Second	64	280
Third	136	377
Fourth	170	475
Fifth	200	510
Sixth	256	646

that they were capable of secreting a range of enzymes and, thus, the strains were presumed to be modified in their secretion mechanisms. It is interesting to note that an improved secreter of a heterologous protein has been isolated using a traditional mutation and selection approach which has been miniaturized and automated.

In several cases the examination of randomly selected high-yielding mutants has indicated that their superiority may have been due to modifications of their control systems. Goulden and Chataway (1969) demonstrated that a mutant of *P. chrysogenum*, producing high levels of penicillin, was less sensitive to the control of acetohydroxyacid synthase by valine. Valine is one of the precursors of penicillin and its synthesis is controlled by its feedback inhibition of acetohydroxyacid synthase. Thus, removal of the control of valine synthesis may result in the production of higher valine levels and, hence, greater production of penicillin. Pruess and Johnson (1967) demonstrated that higher-yielding strains of *P. chrysogenum* also contained higher levels of acyltransferase, the enzyme which catalyses the addition of phenylacetic acid to 6-aminopenicillanic acid. Dulaney (1954) reported that the best producer of streptomycin amongst *Streptomyces griseus* mutants was auxotrophic for vitamin B₁₂ and Demain (1973) claimed that this auxotrophic mutation was still a characteristic of production strains being used in 1969. Thus, it appears that some strains isolated by random selection, and overproducing secondary metabolites, are altered in ways similar to those strains isolated by directed selection techniques and overproducing primary metabolites.

There are many examples in the literature where a more directed selection approach has been adopted for the improvement of secondary metabolite producers.

The techniques used include the isolation of auxotrophs, revertants and analogue-resistant mutants.

THE ISOLATION OF AUXOTROPHIC MUTANTS

Although mutation of secondary metabolite producers to auxotrophy has resulted frequently in their producing lower yields, cases of improved productivity have also been demonstrated. For example, Alikhanian *et al.* (1959) investigated the tetracycline producing abilities of fifty-three auxotrophic mutants, all of which produced significantly less tetracycline than the parent strain in normal production medium. Supplementation of the medium with the growth requirement resulted in one mutant expressing productivity superior to that of the parent.

It is sometimes difficult to explain the precise reason for the effect of mutation to auxotrophy on the production of secondary metabolites, but in the majority of cases it has been demonstrated to be an effect on the secondary metabolic system rather than, simply, an effect on the growth of the organism. The simplest explanation for the deleterious effect on secondary metabolite yield is that the auxotroph is blocked in the biosynthesis of a precursor of the end product, for example, Polsinelli *et al.* (1965) demonstrated that auxotrophs of *Streptomyces antibioticus* which required any of the precursors of actinomycin (isoleucine, valine or threonine) were poor producers of the antibiotic.

Many secondary metabolites may be considered as end products of branched pathways which also give rise to primary metabolites. Thus, a mutation to auxotrophy for the primary end product may also influence the production of the secondary product. In *P. chrysogenum*, lysine and penicillin share the same common biosynthetic route to α -aminoadipic acid, as shown in Fig. 3.25. This biosynthetic route may explain Bonner's (1947) observation that 25% of the lysine auxotrophs of *P. chrysogenum* he isolated could not produce penicillin. The role of lysine in the penicillin fermentation has also led to the investigation of lysine auxotrophs as potential superior penicillin producers. Demain (1957) demonstrated that lysine was inhibitory to the biosynthesis of penicillin. The explanation of this phenomenon is considered to be the inhibition of homocitrate synthase by lysine resulting in the depletion of α -aminoadipic acid required for penicillin synthesis (Demain and Masurekar, 1974). It may be postulated that lysine auxotrophs blocked immediately after α -aminoadipic acid would produce higher levels of penicillin due to the diversion of the intermediate towards

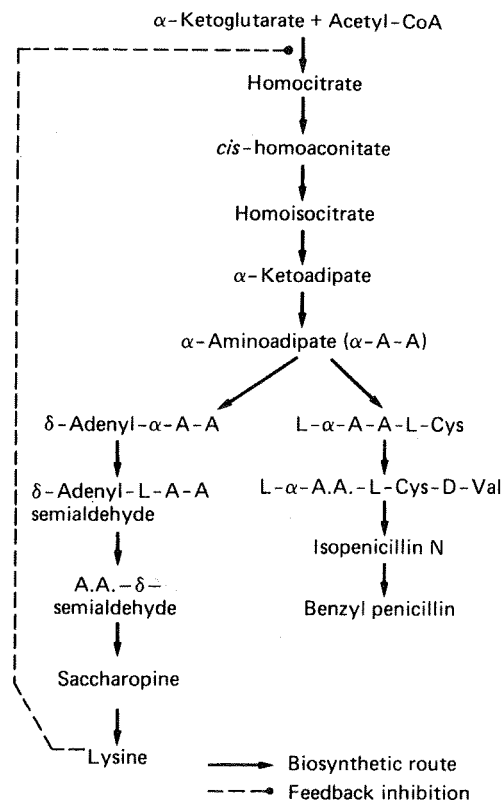


FIG. 3.25. Biosynthesis of benzyl penicillin and lysine in *Penicillium chrysogenum*.

penicillin synthesis and the removal of any control of homocitrate synthase by endogenous lysine. O'Sullivan and Pirt (1973) investigated the production of penicillin by lysine auxotrophs of *P. chrysogenum* in continuous culture but were unable to demonstrate improved productivity in a range of lysine-supplemented media. Luengo *et al.* (1979) examined the lysine regulation of penicillin biosynthesis in low-producing and industrial strains of *P. chrysogenum*. The industrial strain was capable of producing up to 12,000 units cm^{-3} penicillin. These workers demonstrated that although the onset of penicillin synthesis in the industrial strain was less sensitive to lysine than in the low-producing strain, the empirical industrial selection procedures had not completely removed the mechanisms of lysine regulation of penicillin biosynthesis. Luengo *et al.* expressed the possibility that overproducers of penicillin may be obtained by the selection of mutants to lysine regulation. However, it should be noted that the industrial strain employed was a representative of only one series

of penicillin producers and that other series may have already been modified with respect to the regulatory effects of lysine. Nevertheless, this study does indicate the possibility that selection of strains resistant to lysine control may be overproducers of penicillin.

It is far more difficult to explain the effect of auxotrophy for factors not associated with the biosynthesis of the secondary metabolite. Polsinelli *et al.* (1965) demonstrated that seven out of twenty-seven auxotrophs of *S. griseus* produced more actinomycin than did the prototrophic parent. Dulaney and Dulaney (1967) demonstrated increased chlortetracycline yields in eight out of eleven auxotrophs of *S. viridifaciens* when grown in supplemented media. In neither of these cases were the auxotrophic requirements directly involved in the biosynthesis of the secondary metabolites. Demain (1973) put forward two possible explanations to attempt to account for the behaviour of such auxotrophs. The first explanation is that the auxotrophic factors were involved in 'cross-pathway' regulation with the secondary metabolite or its precursors. Demain quoted several examples of cross-pathway regulation in primary metabolism where the activity of one pathway is affected by the product of an apparently unrelated sequence. The alternative explanation is that the effect on secondary metabolism is not due to the auxotrophy but to a second mutation accompanying the auxotrophy, i.e. a double mutation. Demain cited two attempts to determine whether the effects of auxotrophy on secondary metabolism were due to double mutations or to the auxotrophy. MacDonald *et al.* (1963) reverted a low-producing thiosulphate-requiring mutant of *P. chrysogenum* to thiosulphate independence and examined penicillin productivity by the revertants. Approximately half of the revertants re-acquired their 'grandparents' production level, whereas the other half retained their poor productivity. Polsinelli *et al.* (1965) reverted five isoleucine-valine auxotrophs of *S. antibioticus* (which also produced low levels of actinomycin compared with the parent strain) to prototrophy and discovered that some were returned to normal production and others to higher production levels than the 'grandparent'. Thus, in the case of Polsinelli *et al.*'s mutants it is unlikely that the effect on secondary metabolism was due to a double mutation, but it is possible that this was the case for some of MacDonald *et al.*'s strains. However, it should be remembered that both these groups of auxotrophs were poor secondary metabolite producers blocked in routes directly involved with the secondary biosynthetic pathway. It may be more relevant to examine the nature of auxotrophic strains blocked in apparently unrelated pathways and

produce improved levels of the secondary metabolite. It is not possible to say whether any of the auxotrophic mutants previously discussed produced superior levels of the secondary metabolite as a result of double mutations, but it may be possible to exploit this possibility in the future.

The treatment of bacterial cells with nitrosoguanidine (NTG) has been demonstrated to result in clusters of mutations around the replicating fork of the chromosome (Guerola *et al.*, 1971). Thus, if one of the mutations were selectable (for example, auxotrophy) it may be possible to isolate a strain containing the selectable mutation along with other non-selectable mutations which map close by. The efficient application of this technique would depend on the accurate mapping of the gene involved in producing the secondary metabolite so that neighbouring mutations may be selected. This technique may be valuable for the selection of mutants of the bacilli and streptomycetes producing high levels of antibiotics where mutations affecting synthesis may be mapped for each biosynthetic step. Co-mutation by NTG may then be followed by selection for changes to genes adjacent to those loci known to be involved in production of the particular secondary metabolite.

THE ISOLATION OF RESISTANT MUTANTS

The isolation of analogue-resistant mutants has already been discussed in the field of primary metabolism, where the rationale was that a mutant resistant to the inhibitory effects of an analogue, which mimics the control characteristics of the natural metabolite, may overproduce the natural metabolite. This approach has been adopted, or may be adopted, in a number of guises in the field of secondary metabolism.

- (i) Mutants may be isolated which are resistant to the analogues of primary metabolic precursors of the secondary metabolite, thus increasing the availability of the precursor.
- (ii) Mutants may be isolated which are resistant to the feedback effects of the secondary metabolite.
- (iii) Mutants may be selected which are resistant to the toxic effects of the secondary metabolite when added to the trophophase of the producing organism.
- (iv) Mutants may be isolated which are resistant to the toxic effects of a compound due to the production of the secondary metabolite.

(i) *Mutants resistant to the analogues of primary metabolic precursors of the secondary metabolite*

This approach has been adopted by Elander *et al.* (1971) in the isolation of mutants of *Pseudomonas aureofaciens* overproducing the antibiotic pyrrolnitrin. Tryptophan is a precursor of pyrrolnitrin and although it is stimulatory to production it is uneconomic to use as an additive in an industrial process. Thus, Elander *et al.* (1971) isolated mutants resistant to tryptophan analogues using the gradient plate technique described previously. A strain was eventually isolated which produced two to three times more antibiotic than the parent and was resistant to feedback inhibition by tryptophan. Addition of tryptophan to the improved strain would not result in higher pyrrolnitrin synthesis, indicating that tryptophan supply was no longer the limiting factor and the organism was producing sufficient endogenous tryptophan for antibiotic synthesis.

The role of tryptophan in the control of the candicidin fermentation has been investigated by Martin and co-workers (Martin, 1978). Candicidin is a polyene macrolide antibiotic containing an aromatic *p*-aminoacetophenone moiety derived from chorismic acid, a precursor of the aromatic amino acids, so that tryptophan and candicidin may be considered to be end products of a branched biosynthetic pathway. Tryptophan has been demonstrated to inhibit the biosynthesis of the antibiotic which led Martin *et al.* (1979) to isolate mutants resistant to tryptophan analogues. Such mutants produced more candicidin than the parent strain, apparently due to the removal of tryptophan control.

Godfrey (1973) isolated mutants of *Streptomyces lipmanii* resistant to the valine analogue, trifluoroleucine. These mutants produced higher levels of cephamycin than the parent strain, and appeared to be deregulated for the isoleucine, leucine, valine biosynthetic pathway, indicating that valine may have been a rate-limiting step in cephamycin synthesis.

Methionine has been demonstrated to stimulate the biosynthesis of cephalosporin by *Acremonium chrysogenum* and superior producers have been isolated in the form of methionine analogue-resistant mutants (Chang and Elander, 1979). Lysine analogue resistant mutants have yielded a greater frequency of superior β -lactam antibiotic producers compared with random selection (Elander, 1989).

(ii) *Mutants resistant to the feedback effects of the secondary metabolite*

There are many cases in the literature of a secondary metabolite preventing its own synthesis. Martin

(1978) cited the following examples: chloramphenicol, aurodox, cycloheximide, staphylomycin, ristomycin, puromycin, fungicidin, candihexin, mycophenolic acid and penicillin. The precise mechanisms of these controls is not clear but they appear to be specific against the synthesis of the secondary metabolite and not against the general metabolism of the cells. The mechanism of the control of its own synthesis by chloramphenicol appears to be the repression of arylamine synthetase (the first enzyme in the pathway from chorismic acid to chloramphenicol) by chloramphenicol. Jones and Vining (1976) demonstrated that arylamine synthetase was fully repressed by 100 mg dm^{-3} chloramphenicol, a level of antibiotic which neither affected cell growth nor the activities of the other enzymes of the chloramphenicol pathway.

Martin (1978) quoted several examples of correlations between the level of secondary metabolite accumulation and the level which causes 'feedback' control, which may imply that the factor limiting the yields of some secondary metabolites is the feedback inhibition by the end product. The selection of mutants resistant to feedback inhibition by a secondary metabolite is a far more difficult task than the isolation of strains resistant to primary metabolic control. It is extremely unlikely that a toxic analogue of the secondary metabolite could be found where the toxicity lay in the mimicking of the feedback control of the secondary metabolite, a compound which would not be necessary for growth. However, the detection of mutants resistant to feedback inhibition by antibiotics may be achieved by the use of solidified media screening techniques, similar to the miniaturized screening techniques previously described. The technique would involve culturing the survivors of a mutation treatment on solidified medium containing hitherto repressing levels of the antibiotic and detecting improved producers by overlaying the colonies with an indicator organism. The difficulty inherent in this technique is that the incorporated antibiotic, itself, will inhibit the development of the indicator organism. This problem may be overcome by adjusting the depth of the overlay or the concentration of the indicator such that an inhibition zone could be produced only by a level of antibiotic greater than that incorporated in the original medium. However, it is unlikely that this would be a satisfactory solution for the selection of a high-producing commercial strain. Another approach would be to utilize an analogue of the antibiotic which mimicked the feedback control by the natural product but which did not have antimicrobial properties. Inhibition of antibiotic synthesis by analogues has been demonstrated in the

cases of aureodox (Liu *et al.*, 1972) and penicillin (Gordee and Day, 1972). An alternative may be to use a mutant indicator bacterium for the overlay which is only sensitive to levels of the antibiotic in excess of that originally incorporated into the medium. In cases where it has been demonstrated that feedback control by the end product plays an important role in limiting productivity it would probably be worthwhile to design such procedures.

(iii) *The isolation of mutants resistant to the toxic effects of the secondary metabolite in the trophophase*

It has been demonstrated for many secondary metabolite producing organisms that the secondary metabolite is toxic to the producing cell when it is present in the trophophase (growth phase) (Demain, 1974). Thus, it appears that a 'switch' in the metabolism of the organism in the idiophase enables it to produce an otherwise 'autotoxic' product. Furthermore, it has been demonstrated that, in some cases, the higher the resistance to the secondary metabolite in the growth phase the higher is productivity in the production phase. Dolezilova *et al.* (1965) demonstrated that the level of production of nystatin by various strains of *S. noursei* was related to the resistance of the strain to the antibiotic in the growth phase; a non-producing mutant was inhibited by 20 units cm^{-3} , the parent strain produced 6000 units cm^{-3} and was inhibited by 2000 units cm^{-3} in the growth phase and a mutant producing 15,000 units cm^{-3} was found to be resistant to 20,000 units cm^{-3} .

The possible relationship between antibiotic resistance and productivity may be used to advantage in the selection of high-producing mutants by culturing the survivors of a mutation treatment in the presence of a high level of the end product. Those strains capable of growth in the presence of a high level of the antibiotic may also be capable of high productivity in the idiophase. This approach has been used successfully for antifungal agosterols (Bu'Lock, 1980), streptomycin (Woodruff, 1966) and ristomycin (Trenina and Trutneva, 1966) but without success for novobiocin (Hoeksema and Smith, 1961).

A similar rationale was used by McGuire *et al.* (1980) for the improvement of daunorubicin production, an anthracycline antitumour agent synthesized by the red pigmented streptomycete *Streptomyces peuceticus*. The red pigment is presumably anthracycline which shares its oligoketide origin with daunorubicin. The directed selection was based on the ability of the antibiotic cerulenin to suppress oligoketide synthesis. This was indicated in *S. peuceticus* by the lack of the

red pigment on cerulenin agar. Thus, mutants which were still capable of oligoketide synthesis in the presence of the inhibitor would remain red. Approximately a third of the resistant mutants were superior daunorubicin producers.

(iv) *The isolation of mutants in which secondary metabolite synthesis gives resistance to toxic compounds*

A potentially toxic compound may be rendered harmless by an organism converting it to a secondary metabolite or a secondary metabolite complexing with it. Ions of heavy metals such as Hg^{2+} , Cu^{2+} and related organometallic ions are known to complex with β -lactam antibiotics (Elander and Vournakis, 1986) and such agents have been used to select resistant mutants. The logic of this selection process is that overproduction of a β -lactam may be the mechanism for increased resistance to the heavy metal. Elander and Vournakis (1986) reported that the frequency of superior cephalosporin C producers was greater amongst mutants resistant to mercuric chloride than amongst random samples of the survivors of ultraviolet treatment.

The conversion of the penicillin precursor, phenylacetic acid, to penicillin is thought to be an example of detoxification by conversion to a secondary metabolite. It appears that strains capable of withstanding higher concentrations of the precursor may be able to synthesize higher levels of the end product. Poly and Nyiri (1966) applied this hypothesis in selecting phenylacetic acid-resistant mutants of *P. chrysogenum* and demonstrated that 7% of the isolates showed enhanced penicillin production. Barrios-Gonzalez *et al.* (1993) investigated the same phenomenon and developed methods for the enrichment of both spores and early idiophase mycelium resistant to phenyl acetic acid. Of the resistant spore population, 16.7% were superior penicillin producers as compared with 50% of the resistant idiophase population. This suggests that the selective force may be more 'directed' by using a population already committed to penicillin synthesis. Although the best mutants contained elevated levels of acyltransferase (the enzyme that directly detoxifies PAA) they also showed higher levels of isopenicillin N synthetase (cyclase) which the authors claimed to be the limiting step. Thus, it was claimed that the screening method was not specific to selecting for acyltransferase elevation but could select for strains having a faster carbon flow through the pathway enabling them to use PAA faster.

Ball (1978) stressed that the major difficulty in the selection of toxic precursor resistant mutants is that the site of resistance of a mutant may not result in the

organism overproducing the end product — for example, the resistance may be due to an alteration in the permeability of the mutant or due to the mutants' ability to degrade the precursor to a harmless metabolite unrelated to the desired end product. Barrios-Gonzalez's approach of using mycelial fragments already committed to penicillin synthesis decreases the likelihood of such 'false selection' occurring.

THE ISOLATION OF REVERTANT MUTANTS

As discussed in the section on primary metabolites, a mutant may revert to the phenotype of its 'parent', but the genotype of the revertant may not, necessarily, be the same as the original 'parent'. Some revertant auxotrophs have been demonstrated to accumulate primary metabolites (Shiio and Sano, 1969) and attempts have been made to apply the technique to the isolation of mutants overproducing secondary metabolites. Two approaches have been used in the field of secondary metabolites with respect to the isolation of revertants:

- (i) The isolation of revertants of mutants auxotrophic for primary metabolites which may influence the production of a secondary metabolite.
- (ii) The reversion of mutants which have lost the ability to produce the secondary metabolite

(i) *The isolation of revertants of mutants auxotrophic for primary metabolites which may influence the production of a secondary metabolite*

As previously discussed, it is difficult to account for some of the effects of auxotrophy on secondary metabolism and it appears that some may be due to as yet unresolved cross-pathway phenomena and others due to the expression of other mutations associated with the auxotrophy. Similarly, revertant mutants may affect secondary metabolism in a number of ways — direct effects on the pathway, cross-pathway effects and effects due to mutations other than the detected reversion.

Dulaney and Dulaney (1967) investigated the tetracycline productivities of a population of prototroph revertants of *S. viridifaciens* derived from each of five auxotrophs. These workers predicted that some revertants may be productive due to direct influence of the mutations on tetracycline biosynthesis but that others may be so because they contained other lesions. Superior producers were obtained in all the prototroph-re-

vertant populations apart from those derived from a homocysteine auxotroph. However, the frequency of the occurrence of the superior producers was similar to that obtained by the random selection of the survivors of a mutation treatment in all but one of the populations. The exceptional population was the revertants of a methionine auxotroph, 98% of which produced between 1.2 and 3.2 times as much tetracycline as the original prototrophic culture. A possible explanation of the very favourable titres of the population may be the role of methionine as the methyl donor in tetracycline biosynthesis and that methionine availability limited the production of the secondary metabolite in the original prototroph. However, addition of exogenous methionine to the prototroph did not result in superior productivity.

Polsinelli *et al.* (1965) also demonstrated that reversion of five mutants of an actinomycin producing strain of *S. antibioticus* blocked in the isoleucine-valine pathway resulted in the isolation of some superior mutants. Godfrey (1973) reported that the reversion of a cysteine auxotrophic mutant of *S. lipmanii* resulted in improved production of cephamycin.

These studies provide promising evidence that the selection of revertants of auxotrophs of primary metabolites involved in secondary metabolism may yield a high number of productive mutants.

(ii) *The isolation of revertants of mutants which have lost the ability to produce the secondary metabolite*

The reversion of non-producing strains may result in the detection of a high-producing mutant as that mutant would have undergone at least two mutations associated with the production of the secondary metabolite. Dulaney and Dulaney (1967) plated the progeny of a mutation of a non-producing strain of *S. viridifaciens* onto solidified production medium and screened for superior tetracycline producers by an overlay technique. A mutant was isolated which produced nine times the tetracycline yield of the original 'parent'. The major difficulty inherent in this technique is that non-producing mutants of high-yielding strains may be incapable of being reverted due to extreme deficiencies in their metabolism. Indeed, Rowlands (1992) suggested that strains which are non (or low) producers and illustrate other effects such as poor growth and sporulation are best discarded, but that revertant mutants not showing pleiotropic effects are perhaps more likely to possess genuine increases in biosynthetic activity than those produced by any other technique, having been mutated twice in genes directly affecting product formation.

The use of recombination systems for the improvement of industrial micro-organisms

Hopwood (1979) defined recombination, in its broadest sense, as "any process which helps to generate new combinations of genes that were originally present in different individuals". The use of recombination mechanisms for the improvement of industrial strains has increased significantly due to the developments in recombinant DNA technology and the necessity to develop new methods of strain improvement as the returns generated from mutation and selection programmes decreased. However, it should be appreciated that mutation and selection techniques are frequently used in association with recombination systems in a strain improvement programme. The parasexual cycle in the filamentous fungi has been applied to strain development as have protoplast fusion techniques in a wide range of micro-organisms.

THE APPLICATION OF THE PARASEXUAL CYCLE

Many industrially important fungi do not possess a sexual stage and therefore it would appear difficult to achieve recombination in these organisms. However, Pontecorvo *et al.* (1953) demonstrated that nuclear fusion and gene segregation could take place outside, or in the absence of, the sexual organs. The process was termed the parasexual cycle and has been demonstrated in the imperfect fungi, *A. niger* and *P. chrysogenum*, as well as the sexual fungus *A. nidulans*. In order for parasexual recombination to take place in an imperfect fungus, nuclear fusion must occur between unlike nuclei in the vegetative hyphae of the organism. Thus, recombination may be achieved only in an organism in which at least two different types of nuclei coexist, i.e. a heterokaryon. The heterozygous diploid nucleus resulting from the fusion of the two different haploid nuclei may give rise to a diploid clone and, in rare cases, a diploid nucleus in the clone may undergo an abnormal mitosis resulting in mitotic segregation and the development of recombinant clones which may be either diploid or haploid.

Recombinant clones may be detected by their display of recessive characteristics not expressed in the heterokaryon. Analysis of the recombinants normally demonstrates them to be segregant for only one, or a few linked, markers and culture of the segregants results in the development of clones displaying more recessive characters than the initial segregant. The process of recombination during the growth of the

heterozygous diploid may occur in two ways: mitotic crossing over, which results in diploid recombinants, and haploidization, which results in haploid recombinants.

Mitotic crossing over is the result of an abnormal mitosis. The normal mitosis of a heterozygous diploid cell is shown in Fig. 3.26. During mitosis, each pair of homologous chromosomes replicate to produce two pairs of chromatids and a chromatid of one pair migrates to a pole of the cell with a chromatid of the other pair. Division of the cell at the equator results in the production of two cells, both of which are heterozygous for all the genes on the chromosome. Mitotic crossing over involves the exchange of distal segments between chromatids of homologous chromosomes as shown in Fig. 3.27. This process may result in the production of daughter nuclei homozygous for a portion of one pair of chromosomes and in the expression of any recessive alleles contained in that portion. Thus, the clone arising from the partial homozygote will be recombinant and further mitotic crossing over in the recombinant will result in the expression of more recessive alleles.

Haploidization is a process which results in the unequal distribution of chromatids between the progeny of a mitosis. Thus, of the four chromatids of an homologous chromosome pair, three may migrate to one pole and one to another resulting in the formation of two nuclei, one containing $2n + 1$ chromosomes and the

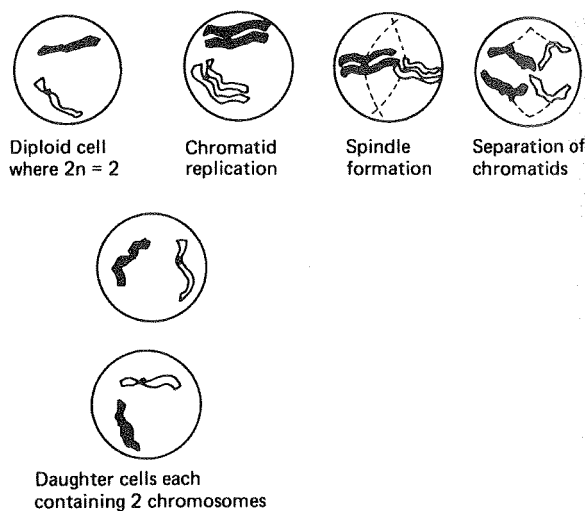


FIG. 3.26. Diagrammatic representation of the mitotic division of a eukaryotic cell containing two chromosomes. The nuclear membrane has not been portrayed in the figure.

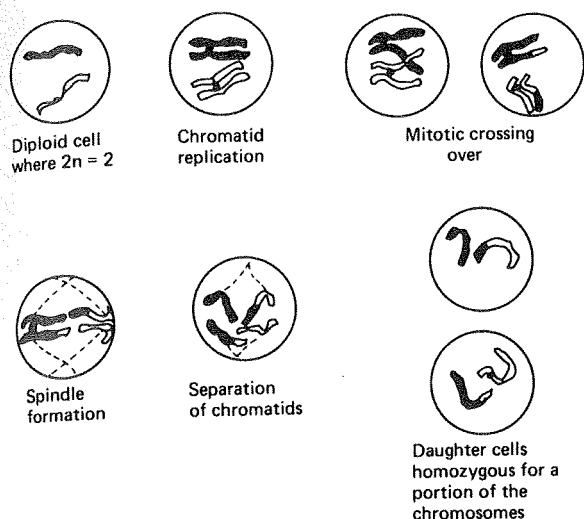


FIG. 3.27. Diagrammatic representation of mitosis including mitotic crossing over.

other containing $2n - 1$ chromosomes. The $2n - 1$ nuclei tend towards the haploid state by the progressive random loss of further chromosomes. Thus, the resulting haploid nucleus will contain a random assortment of the homologues of the chromosomes of the organism. A representation of the process is shown in Fig. 3.28.

Therefore, the major components of the parasexual cycle are the establishment of a heterokaryon, vegetative nuclear fusion and mitotic crossing over or haploidization resulting in the formation of a recombinant. In practice, the occurrence and detection of these stages may be enhanced by the use of auxotrophic markers. The parents of the cross are made auxotrophic for different requirements and cultured together on minimal medium. The auxotrophs will grow very slightly due to the carry over of their growth requirements from the previous media, but if a heterokaryon is produced, by anastomoses forming between the two parents, then it will grow rapidly. The frequency of vegetative nuclear fusion in the heterokaryon may be enhanced by the use of agents such as camphor vapour or ultraviolet light; mitotic segregation may be enhanced by the use of agents such as X-rays, nitrogen mustard, *p*-fluorophenylalanine and ultraviolet light (Sermonti, 1969).

The application of the parasexual cycle to industrially important fungi has been hindered by a number of problems (Elander, 1980). A major difficulty is the influence of the auxotrophic markers (used for the selection of the heterokaryon) on the synthesis of the

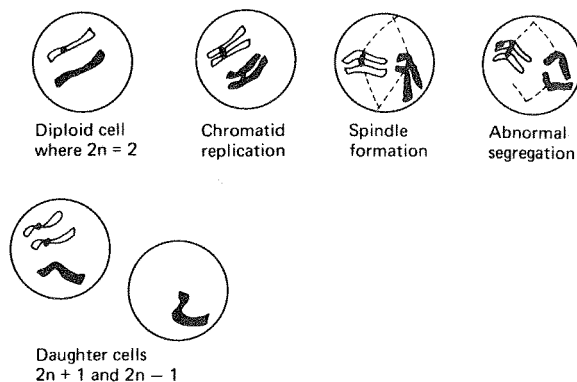


FIG. 3.28. Diagrammatic representation of mitosis involving haploidization.

desired product. As discussed earlier, auxotrophic mutations have been observed to have quite unpredictable results on the production of some secondary metabolites. Even conidial colour markers have been shown to have deleterious effects on product synthesis (Elander, 1980). Even when suitable markers are available, the induction of heterokaryons in some industrial fungi has been demonstrated to be a difficult process and specialized techniques had to be employed to increase the probabilities of heterokaryon formation (see MacDonald and Holt, 1976). However, the development of protoplast fusion methods in the late 1970s enabled efficient heterokaryon formation to be achieved and removed the major barrier to the application of the parasexual cycle to strain improvement. These developments are considered in more detail in the next section on protoplast fusion.

Despite the early difficulties of inducing the parasexual cycle in some industrial fungi it was used in two ways to study these organisms. The cycle was used to investigate the genetics of the producing strains as well as to develop recombinant superior producers. Information obtained on the basic genetics of the industrial fungi using the parasexual cycle included the number of chromosomes (or linkage groups), the allocation of genes, important in product synthesis, to particular chromosomes and the mapping of important genes on a chromosome. Sermonti (1969) and MacDonald and Holt (1976) have described the techniques which were used to achieve these objectives.

Initial studies on haploid strains of *P. chrysogenum* derived from parasexual crosses demonstrated that most of the progeny exhibited the genotype of one of the parents, i.e. no recombination had occurred (MacDonald, 1968). MacDonald suggested that one of the

reasons for this lack of success may have been the differences in gross chromosomal morphology between the parents caused by certain mutagens used in the development of the strains. Thus, one precaution to be adopted in the development of strains which may be used in a parasexual cross is the avoidance of mutagens which may cause gross changes in chromosomal morphology. Subsequently, Ball (1978) suggested that the careful choice of markers and the use of strains giving similar titres and being 'not too divergent' may result in achieving more recombinants.

Although diploids produced by the parasexual cycle are frequently unstable, stable diploids have been reported to have been used for the industrial production of penicillin. Elander (1967) isolated a diploid from a sister cross of *P. chrysogenum* which was shown to be a better penicillin producer than the parents and was morphologically more stable (a sister cross is one where the two strains differ only in the markers used to induce the heterokaryon). One explanation of the superior performance of the diploid may have been its resistance to strain degeneration caused by deleterious recessive mutations. Such mutations would only have been expressed in the diploid if both alleles had been modified. Calam *et al.* (1976) demonstrated that a diploid strain of *P. chrysogenum* was more stable than haploid mutants and mutation and selection of the diploid gave rise to a diploid strain producing higher levels of penicillin than the parents. Ball (1978) claimed that the usefulness of diploids may only be short term, presumably implying problems of degeneration, and may best be used as stepping stones to a recombinant haploid.

The advantages to be gained from the industrial use of parasexual recombinants are not confined to the amalgamation of different yield improving mutations. Equally advantageous would be the introduction of characteristics which would make the process more economic, for example low viscosity, sporulation and the elimination of unwanted products. The development of these issues will be considered in the next section on protoplast fusion.

THE APPLICATION OF PROTOPLAST FUSION TECHNIQUES

Protoplasts are cells devoid of their cell walls and may be prepared by subjecting cells to the action of wall degrading enzymes in isotonic solutions. Protoplasts may regenerate their cell walls and are then capable of growth as normal cells. Cell fusion, followed by nuclear fusion, may occur between protoplasts of

strains which would otherwise not fuse and the resulting fused protoplast may regenerate a cell wall and grow as a normal cell. Thus, protoplasts may be used to overcome some recombination barriers. Protoplast fusion has been demonstrated in a large number of industrially important organisms including *Streptomyces* spp. (Hopwood *et al.*, 1977), *Bacillus* spp. (Fodor and Alfoldi, 1976), corynebacteria (Karasawa *et al.*, 1986), filamentous fungi (Ferenczy *et al.*, 1974) and yeasts (Sipiczki and Ferenczy, 1977).

Fusion of fungal protoplasts appears to be an excellent technique to obtain heterokaryons between strains where conventional techniques have failed, or, indeed, as the method of choice. Thus, this approach has allowed the use of the parasexual cycle for breeding purposes in situations where it had not been previously possible. This situation is illustrated by the work of Peberdy *et al.* (1977) who succeeded in obtaining heterokaryons between *P. chrysogenum* and *P. cyaneofulvum* and demonstrated the formation of diploids which gave rise to recombinants after treatment with *p*-fluorophenylalanine or benomyl. Although it has been claimed that *P. chrysogenum* and *P. cyaneofulvum* are not different species of *Penicillium* (Samson *et al.*, 1977), Peberdy *et al.* still demonstrated that protoplast fusion could be successful where conventional techniques had failed.

A demonstration of the use of protoplast fusion for an industrial fungus is provided by the work of Hamlyn and Ball (1979) on the cephalosporin producer, *C. acremonium*. These workers compared the effectiveness of conventional techniques of obtaining nuclear fusion between strains of *C. acremonium* with the protoplast fusion technique. The results from conventional techniques suggested that nuclear fusion was difficult to achieve. Electron microscopic examination of fused protoplasts indicated that up to 1% underwent immediate nuclear fusion. Recombinants were obtained in both sister and divergent crosses. A cross between an asporulating, slow-growing strain with a sporulating fast-growing strain which only produced one-third of the cephalosporin level of the first strain eventually resulted in the isolation of a recombinant which combined the desirable properties of both strains, i.e. a strain which demonstrated good sporulation, a high growth rate and produced 40% more antibiotic than the higher-yielding parent. Chang *et al.* (1982) utilized protoplast fusion to combine the desirable qualities of two strains of *Penicillium chrysogenum*. Protoplasts from two strains, differing in colony morphology and the abilities to produce penicillin V (the desired product)

and OH-V penicillin (an undesirable product), were fused, followed by plating on a non-selective medium. Out of 100 stable colonies which were scored, two possessed the desirable morphology, high penicillin V and low OH-V penicillin productivities.

Lein (1986) reported the penicillin strain improvement programme adopted by Panlabs, Inc. This programme included random and directed selection as well as protoplast fusion. Table 3.7 illustrates the properties of the two strains used in a protoplast fusion and one of the recombinants selected. To avoid any adverse effects no selective genetic markers were used and the regenerated colonies were screened on the basis of colony morphology and spore colour. A total of 238 colonies judged to be recombinants were screened for penicillin V production and the culture with the best combination of properties is shown in the table. Thus, the desirable characteristics of each strain were combined in the recombinant.

Protoplasts are also useful in the filamentous fungi for manipulations other than cell fusion. Rowlands (1992) suggested that they may be used in mutagenesis of non-sporulating fungi. Spores are the cells of choice for the mutagenesis of filamentous fungi but this is obviously impossible for non-sporulating strains. Mycelial fragments may be used but these will be multinucleate and very high mutagen doses are required. Although some protoplasts will be non-nucleate or multi-nucleate at least some will be uninucleate which will express any modified genes after mutation. Also, protoplasts will take up DNA in *in vitro* genetic manipulation experiments and this aspect will be discussed in a later section of this chapter.

Recombination can take place between actinomycetes by conjugation (Hopwood, 1976) and phage transduction (Studdard, 1979). However, both these mechanisms involve the transfer of only small regions of the bacterial chromosome. Furthermore, the low frequencies at which recombination occurs necessitate the use of selectable genetic markers such as auxotro-

phy or antibiotic resistance. The introduction of such markers is time-consuming but also they can detrimentally affect the synthetic capacity of the organism. Protoplast fusion has particular advantages over conjugation in that the technique involves the participation of the entire genome in recombination. Also, Hopwood (1979) has developed techniques which have resulted in the recovery of a very high proportion of recombinants from the fusion products of *Streptomyces coelicolor* protoplasts. By subjecting protoplasts to an exposure of ultraviolet light, sufficient to kill 99% of them prior to fusion, Hopwood has claimed a tenfold increase in recombinant detection for strains normally giving a low yield of recombinants (1%) and a doubling of the recombination frequency for a cross normally yielding 20% recombinants. Such yields of recombinants means that they would be detectable by simply screening a random proportion of the progeny of a protoplast fusion and the use of selectable markers to 'force' out the recombinants would not be necessary. Examples of the application of the technique to actinomycete strain improvement include cephamycin C yield enhancement in *Nocardia lactamdurans* (Wesseling, 1982) and the improvement of lignin degradation in *Streptomyces viridosporus* (Petty and Crawford, 1984).

Protoplast fusion has also been applied to the improvement of amino acid producing strains. Karasawa *et al.* (1986) used the technique to improve the fermentation rates of lysine producers developed using repeated mutation and directed selection. Such strains were good lysine producers but showed low glucose consumption and growth rates, undesirable features which had been inadvertently introduced during the selection programme. A protoplast fusion was performed between the lysine producer and a fast growing strain and a fusant was isolated displaying the desirable characteristics of high lysine production and high glucose consumption rate resulting in a much faster fermentation. The same authors used protoplast fusion to produce a superior threonine producing *Brevibacterium*

TABLE 3.7. The use of protoplast fusion for the improvement of a penicillin V producer (Lein, 1986)

Characteristic	Parent A	Parent B	Best recombinant
Spores per slant ($\times 10^8$)	2.2	2.5	7.5
Germination frequency (%)	99	40	49
Colour of sporulating colonies	Green	Pale green	Deep green
Seed growth	Good	Poor	Good
Penicillin V yield (mg cm ⁻³)	11.7	18.5	18.0
Phenylacetic oxidation	Yes	No	No

lactofermentum strain. Lysine auxotrophy was introduced into a threonine and lysine overproducer by fusing it with a lysine auxotroph — the recombinant produced higher levels of threonine due to its lysine auxotrophy.

THE APPLICATION OF RECOMBINANT DNA TECHNIQUES

The transfer of DNA between different species of bacteria has been achieved experimentally using both *in vivo* and *in vitro* techniques (Atherton *et al.*, 1979). Thus, genetic material derived from one species may be incorporated into another where it may be expressed. *In vivo* techniques make use of phage particles which will pick up genetic information from the chromosome of one bacterial species, infect another bacterial species and in so doing introduce the genetic information from the first host. The information from the first host may then be expressed in the second host. Whereas, the *in vivo* techniques depend on vectors collecting information from one cell and incorporating it into another, the *in vitro* techniques involve the insertion of the information into the vector by *in vitro* manipulation followed by the insertion of the carrier and its associated 'extra' DNA into the recipient cell. Because the DNA is incorporated into the vector by *in vitro* methods the source of the DNA is not limited to that of the host organism of the vector. Thus, DNA from human or animal cells may be introduced into the recipient cell. Atherton *et al.* (1979) listed the basic requirements for the *in vitro* transfer and expression of foreign DNA in a host micro-organism as follows:

- (i) A 'vector' DNA molecule (plasmid or phage) capable of entering the host cell and replicating within it. Ideally the vector should be small, easily prepared and must contain at least one site where integration of foreign DNA will not destroy an essential function.
- (ii) A method of splicing foreign genetic information into the vector.
- (iii) A method of introducing the vector/foreign DNA recombinants into the host cell and selecting for their presence. Commonly used simple characteristics include drug resistance, immunity, plaque formation, or an inserted gene recognizable by its ability to complement a known auxotroph.
- (iv) A method of assaying for the 'foreign' gene product of choice from the population of recombinants created.

The initial work focused on *E. coli* but subsequently techniques have been developed for the insertion of foreign DNA into a range of bacteria, yeasts, filamentous fungi and animal cells. The range of vectors has been discussed by Gingold (1993) for bacteria, Curran and Bugeja (1993) for yeasts, Hopwood *et al.* (1985a) for streptomycetes, Elander (1989) for filamentous fungi and Murray (1993) for animal cells. The insertion of information into the vector molecule is achieved by the action of restriction endonucleases and DNA ligase. Site-specific endonucleases produce specific DNA fragments which may be joined to another similarly treated DNA molecule using DNA ligase. The modified vector is then normally introduced into the recipient cell by transformation. Because the transformation process is an inefficient one, selectable genes must be incorporated into the vector DNA so that the transformed cells may be cultured preferentially from the mixture of transformed and parental cells. This is normally accomplished by the use of drug-resistant markers so that those cells containing the vector will be capable of growth in the presence of a certain antimicrobial agent. The process is shown diagrammatically in Fig. 3.29.

Once the desired gene has been introduced into the recipient cell the problem of expression of the gene arises. This is particularly difficult when a eukaryotic gene is introduced into a bacterium. In the late 1970s a large number of mammalian genes were successfully introduced into bacterial cells, but there was little evidence of any gene expression (Atherton *et al.*, 1979). The problem of eukaryotic gene expression in prokaryotic cells is due to the different structure of eukaryote genes which contain non-coding segments of DNA. Thus, the production of a eukaryotic product by a prokaryotic cell necessitates the incorporation of the genes coding for the product in a form that may be translated by the recipient cell. Two approaches have been adopted to construct eukaryotic genes in a prokaryotic form: the first is to synthesize DNA corresponding to the primary structure of the protein product of the gene, although this method is suitable only for the construction of genes coding for small peptides of known structure. Itakura *et al.* (1977) synthesized the gene coding for the human hormone, somatostatin, and succeeded in incorporating it in *E. coli* where it was expressed. The alternative technique is to synthesize DNA from the messenger RNA, corresponding to the gene, using the enzyme, reverse transcriptase. Eukaryote messenger RNA is similar to bacterial RNA in that it does not contain non-coding segments so that DNA synthesized from an eukaryotic messenger RNA template should be in a form which is transcribable by a

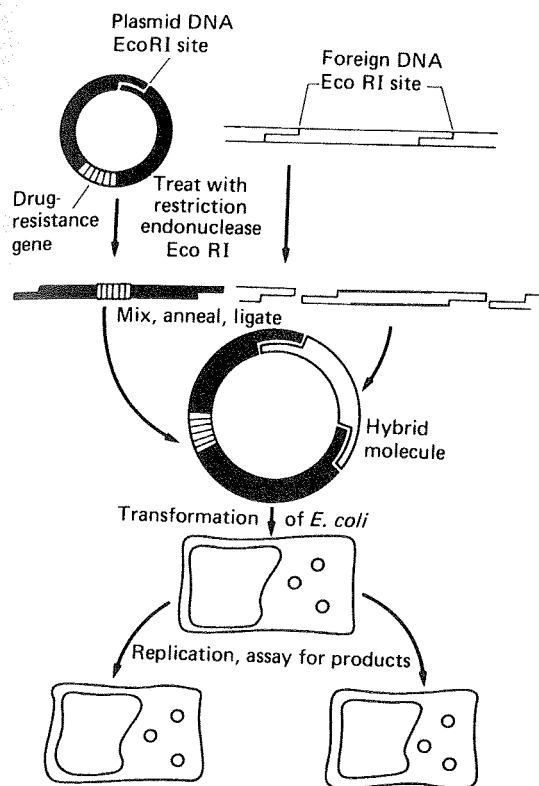


FIG. 3.29. A summary of the steps in *in vitro* genetic recombination. Both plasmid vector and foreign DNA are cut by the restriction endonuclease, EcoRI, producing linear double-stranded DNA fragments with single-stranded cohesive projections. EcoRI recognizes the oligonucleotide sequence $\begin{smallmatrix} \text{GAATTC} \\ \text{CTTAAG} \end{smallmatrix}$ and will cut any double-stranded DNA molecule to yield fragments with the same cohesive ends $\begin{smallmatrix} \text{GAATTC} \\ \text{CTTAAG} \end{smallmatrix}$. On mixing vector and foreign DNA, hybrids form into circular molecules which can be covalently joined using DNA ligase. Transformation of *E. coli* results in the low-frequency uptake of hybrid molecules whose presence can be detected by the ability of the plasmid to confer drug resistance on the host (Atherton *et al.*, 1979).

prokaryotic cell. Nagata *et al.* (1980) used the reverse transcriptase method to produce the genes coding for human interferon. Complementary single-stranded DNA was prepared from a mixture of messenger RNA extracted from virus-induced human lymphocytes. The DNA was then introduced into *E. coli* at random using a plasmid vector and the recipient cells screened for the presence of the interferon gene. Those colonies that contained the gene were then examined for interferon production. Using this method, Nagata's group succeeded in demonstrating the expression of the interferon gene in *E. coli*. Regrettably, a detailed explanation of the techniques involved in gene cloning is

outside the scope of this book, but excellent accounts of the various methods are given by Gingold (1993), Slater (1993) and Curran and Bugeja (1993), all in the same text (Walker and Gingold, 1993).

The most publicized application of recombinant DNA technology in the context of fermentation is the construction of strains capable of synthesizing foreign proteins. Although this chapter is concerned with strain improvement it is the obvious place to consider such chimeric strains. The use of the technique for the improvement of microbial product synthesis has also been very successful and this will be discussed in a later section.

(i) The production of heterologous proteins

The rationale for the commercial production of foreign proteins in micro-organisms depends on the protein under consideration. The first commercial heterologous protein to be produced was human growth hormone (hGH) which is used to treat hypopituitary dwarfism and, prior to its manufacture by fermentation, was extracted from the brains of human cadavers. Naturally, this source was not readily available and carried the additional disadvantage of the risk of contamination with human pathogens. The successful production of recombinant hGH from *E. coli* both satisfied the demand for the compound and eliminated the risks associated with the human source (Dykes, 1993). Factor VIII is a blood clotting agent used in the treatment of haemophilia. Prior to its production as a recombinant protein it was extracted from human blood with the associated risk of contamination with HIV, so that the logic behind its production by fermentation is very similar to that of hGH — availability and safety.

The logic for the development of recombinant human insulin is not quite as clear because diabetes had been treated successfully for many years with animal insulin. However, it was assumed that the recombinant human product would cause fewer immunological difficulties and it would be pure and not contaminated with such pancreatic peptides as proinsulin, glucagon, somatostatin, pancreatic polypeptides and vaso-active intestinal peptides. Furthermore, the incidence of diabetes was also expected to increase due to changes in diet, the improved care of pregnant diabetic women (resulting in an increase of diabetes in the gene pool) and the increased life expectancy of diabetics (Dykes, 1993). Recombinant human insulin was first marketed in the U.K. in 1982 and by 1989 had become the most common form in use (Dykes, 1993).

Many other human proteins are synthesized at very low levels and the only practical way to produce them

in sufficient quantities for use as therapeutic agents is as recombinant proteins. Examples of such products include the interferons and erythropoietin. Table 3.8 lists the recombinant proteins which have been licensed for therapeutic use. According to Dykes (1993) although insulin and hGH have been two of the most successful products (in terms of sales) some newer products have greater potential. For example, erythropoietin stimulates the production of erythrocytes from immature erythroid progenitor cells and has been used for the successful treatment of renal failure-induced anaemia. Predicted annual sales of the protein in the mid-1990s are in the region of \$1200 million. The potential markets of some other recombinant human proteins are considered in Chapter 1.

Hepatitis B virus causes an infection of the liver giving rise to chronic viral hepatitis which may lead to progressive liver disease. There is no effective cure of the disease and, thus, vaccination is critical. The first hepatitis B vaccine became available in 1982, prepared from hepatitis B surface antigen (HBsAg) purified from the plasma of human carriers of the disease. Although the vaccine was successful its origin obviously presented the same difficulties of availability and contamination as discussed for hGH and Factor VIII. These difficulties provided the impetus to develop a recombinant vaccine which was achieved by expressing the HBsAg gene in *Saccharomyces cerevisiae*.

The vast majority of the early work on recombinant DNA technology concerned the transfer of genetic material into *E. coli* and this led to the adoption of the organism as the host for the production of several heterologous proteins. The bacterium had several advantages in its favour, primarily:

The availability of genetic knowledge.

The very wide range of vectors available.

A simple fermentation process using cheap media. Promising protein yields in the range of 2–5 g dm⁻³.

Despite these advantages *E. coli* also presents several problems for the production of heterologous proteins:

Proteins are formed as insoluble aggregates.

The proteins are not secreted.

Lack of post-translational modification.

By the mid-1980s the use of *E. coli* had declined considerably and yeasts, filamentous fungi and animal cells in culture were being investigated as alternative hosts. However, by 1992 *E. coli* was once again in favour. The protein yields obtained from animal cells were disappointing and the understanding of secretion and protein folding had progressed such that soluble proteins could be secreted by engineered *E. coli* (Hockney, 1994). However, mammalian cells are still the preferred host when the protein activity depends upon post-translational modification.

Whichever organism is used for the synthesis of an heterologous protein it is important that the strain is constructed such that expression of the gene may be controlled during the fermentation process. This may be achieved by inserting the gene into an inducible system, such as β -galactosidase, so that expression of the gene may be initiated by the addition of the inducer. This aspect is discussed in Chapter 4.

As an example of the production of an heterologous protein the development of the Genencor process for the production of recombinant chymosin will be con-

TABLE 3.8. Recombinant proteins licensed for therapeutic use (Dykes, 1993)

Protein	Clinical use
Insulin	Diabetes
Growth hormone	Hypopituitary dwarfism
Tissue plasminogen activator	Clot lysis
Erythropoietin	Anaemia
G-CSF	Cancer chemotherapy
GM-CSF	Bone marrow transplantation
Factor VIII	Haemophilia
Interferon α	Cancers, hepatitis B, leukaemia
Interferon β	Cancers, amyotrophic lateral sclerosis, genital warts
Interferon	Cancers, AIDS-related complex, osteopetrosis
Hepatitis B surface antigen	Hepatitis vaccine

sidered (Ward *et al.*, 1990; Dunn-Coleman *et al.*, 1991). Bovine chymosin is an aspartyl protease extracted from the abomasum of unweaned calves and is used as a milk-clotting agent in the manufacture of cheese. Its natural source means that the enzyme is available only in limited quantities which made the possibility of producing it as an heterologous protein an attractive proposition. This process utilizes the filamentous fungus *Aspergillus niger* var. *awamori* which was claimed to have the following advantages as a potential host organism:

It is capable of secreting large amounts of protein.
The organism is regarded as safe.
Transformation systems were available.

Secretion from a eukaryotic cell is achieved via the endoplasmic reticulum and the Golgi apparatus. Entry into the secretory pathway is determined by the presence of a short, hydrophobic 'signal' sequence on the *N*-terminal end of secreted proteins. The signal sequence directs the protein to sites on the endoplasmic reticulum (ER) from where transport into the lumen of the ER occurs. The signal sequence is cleaved by a luminal signal peptidase and the protein passes through the ER and Golgi body, before being packaged into secretory vesicles and exported beyond the cell membrane (Curran and Bugeja, 1993).

Several expression vectors were constructed incorporating the organism's glucoamylase gene and prochymosin cDNA. The inclusion of the prochymosin cDNA between the glucoamylase promoter and terminator resulted in a strain which accumulated only 15 mg dm⁻³ chymosin in the medium despite high intracellular mRNA and chymosin levels. This implied that the enzyme was not being secreted. To facilitate secretion, the prochymosin cDNA was incorporated between the glucoamylase coding region and the glucoamylase terminator. In this case, the organism should synthesize a 'fusion protein' consisting of chymosin attached to the carboxyl terminus of glucoamylase. The fusion protein was secreted and cleaved autocatalytically liberating the chymosin which was produced at 150 mg dm⁻³.

It is fascinating to appreciate that further improvement of this strain has been achieved by mutant induction and selection. Mutagenesis followed by random selection resulted in the isolation of a superior producer which was subsequently shown to produce very low amounts of the native aspartyl protease. Armed with this information the Genencor group cloned the aspartyl protease and then deleted the gene from the strain. Thus, the insight gained from the random screen

allowed the directed *in vitro* approach to be used. The improved strain produced 250 mg dm⁻³ chymosin. This strain was then used in further mutation and selection programmes. An automated microtitre plate screen was developed which has already been described in an earlier section of this chapter and the results are shown in Table 3.6.

The strains developed using the automated screen were then mutated and subjected to a directed screen. Spores which survived nitrosoguanidine exposure were plated onto glycerol media containing 1% deoxyglucose. Deoxyglucose is a toxic compound which Allen *et al.* (1989) used to isolate resistant mutants of *Neurospora crassa* capable of producing higher levels of extracellular enzymes normally repressed by glucose. The best resistant strains (termed dgr) produced in excess of 1200 mg dm⁻³. Parasexual crosses between improved strains showed that two different loci were involved in resistance to deoxyglucose.

The chymosin story is an excellent example of the integration of recombinant DNA technology, mutant induction and automated random selection, directed selection and parasexual genetics in the development of a commercial strain.

(ii) The use of recombinant DNA technology for the improvement of native microbial products

Recombinant DNA technology has been used widely for the improvement of native microbial products. Frequently, this has involved 'self cloning' work where a chromosomal gene is inserted into a plasmid and the plasmid incorporated into the original strain and maintained at a high copy number. Thus, this is not an example of recombination because the engineered strain is altered only in the number of copies of the gene and does not contain genes which were present originally in a different organism. However, the techniques employed in the construction of these strains are the same as those used in the construction of chimeric strains, so it is logical to consider this aspect here.

The first application of gene amplification to industrial strains was for the improvement of enzyme production. Indeed, some regulatory mutants isolated by conventional means owed their productivity to their containing multiple copies of the relevant gene as well as the regulatory lesion. For example, the *E. coli* β -galactosidase constitutive mutants isolated by Horiuchi *et al.* (1963) in chemostat culture also contained up to four copies of the *lacZ* gene. According to Demain (1990), during the 1960s and early 1970s the number of gene copies were increased by using plasmids or trans-

ducing phage in the same species. The production of β -galactosidase, penicillinase, chloramphenicol transacetylase and aspartate transcarbamylase were all increased by transferring plasmids containing the structural gene into recipient cultures, especially when the plasmid replicated faster than the host chromosome. The advent of recombinant DNA technology increased the applicability of this approach by allowing the construction of vectors containing the desired gene and enabling the transfer of DNA to other species. Table 3.9 includes examples of gene amplification giving rise to improved enzyme yields.

The development of the process for the production of a lipase for use in washing powders by Novo Industri is an excellent example of the application of recombinant DNA technology to enzyme fermentations, although the technical details of the exercise are unavailable (Upshall, 1992). A lipase was isolated from a fungus which was unsuitable for commercial development. A cDNA clone coding for the lipase was prepared and transformed into an industrial *Aspergillus* strain used for enzyme production. The recombinant *Aspergillus* produced the lipase at high levels. Only 8 months elapsed between cloning the gene and the first commercial fermentation. Thus, recombinant DNA technology allowed the exploitation of a valuable microbial enzyme by facilitating its production in a well-established commercially acceptable strain.

The first successful application of genetic engineering techniques to the production of amino acids was obtained in threonine production with *E. coli*. Debabov (1982) investigated the production of threonine by a threonine analogue resistant mutant of *E. coli* K12. The entire threonine operon was introduced into a plasmid which was then incorporated into the organism by transformation. The plasmid copy number in the cell was approximately twenty and the activity of the threonine operon enzymes (measured as homoserine dehy-

drogenase activity) was increased 40–50 times. The manipulated organism produced 30 g dm^{-3} threonine, compared with $2\text{--}3 \text{ g dm}^{-3}$ by the non-manipulated strain. Miwa *et al.* (1983) utilized similar techniques in constructing an *E. coli* strain capable of synthesizing 65 g dm^{-3} threonine. It is important to appreciate that the genes which were amplified in these production strains were already resistant to feedback repression so that the multi-copies present in the modified organism were expressed and not subject to control. Thus, the recombinant DNA techniques built on the achievements made with directed mutant isolation.

The application of genetic engineering to the industrially important corynebacteria was hampered for some years by the lack of suitable vectors. However, vectors have been constructed from corynebacterial plasmids and transformation and selective systems developed. The first patents for suitable vectors were registered by the two Japanese companies Ajinomoto (1983) and Kyowa Hakko Kogyo (1983) and now a range of vectors is available with kanamycin, chloramphenicol and hygromycin as common selectable resistance markers (Martin, 1989). Transformation has been achieved using protoplasts and, more recently, electroporation has been used successfully to introduce the required DNA (Dunican and Shivan, 1989). These systems have enabled not only the use of recombinant DNA technology for strain improvement but have also facilitated the detailed investigation of the molecular biology of these important amino acid and nucleotide producers; for example the molecular organization of the pathway to the aspartate family of amino acids in *C. glutamicum* has been elucidated.

It is not surprising that the improvement of threonine production was the first reported use of recombinant DNA technology with amino acid producing corynebacteria (Shiio and Nakamori, 1989). It may be recalled from the discussion of the development of the

TABLE 3.9. Examples of the enhancement of enzyme production by gene amplification (After Demain, 1990)

Enzyme	Gene donor	Gene recipient	Increase (fold)	Reference
α -Amylase	<i>Bacillus amyloliquefaciens</i>	<i>Bacillus subtilis</i>	10	Sibakov <i>et al.</i> (1983)
α -Amylase	<i>Bacillus stearothermophilus</i>	<i>Bacillus stearothermophilus</i>	5	Aiba <i>et al.</i> (1983)
α -Amylase	<i>Bacillus stearothermophilus</i>	<i>Bacillus brevis</i>	100	Tsukagoshi <i>et al.</i> (1985)
EcoRI restriction endonuclease	<i>E. coli</i>	<i>E. coli</i>	50–100	Cheng <i>et al.</i> (1984)
DNA polymerase	<i>E. coli</i>	<i>E. coli</i>	100	Kelley <i>et al.</i> (1987)

early threonine producers that *C. glutamicum* was not particularly amenable for threonine over-production using auxotrophs and analogue resistant mutants. Over-production was achieved by incorporating a DNA fragment coding for homoserine dehydrogenase from a *Brevibacterium lactofermentum* threonine producer into a plasmid and introducing the modified plasmid back into the *Brevibacterium*. A similar approach was used for homoserine kinase and a strain was developed with remarkably increased homoserine kinase and dehydrogenase activities which produced 33 g dm^{-3} threonine (Morinaga *et al.*, 1987).

The application of recombinant DNA technology to the development of processes for the production of phenylalanine is an excellent illustration of the inter-relationship between mutant development and genetic engineering. Phenylalanine is a precursor of the sweetener, aspartame, and is thus an exceptionally important fermentation product. Backman *et al.* (1990) described the rationale used in the construction of an *E. coli* strain capable of synthesizing commercial levels of phenylalanine. *E. coli* was chosen as the producer because of its rapid growth, the availability of recombinant DNA techniques and the extensive genetic database.

The control of the biosynthesis of the aromatic family of amino acids in *E. coli* is shown in Fig. 3.30. The first step in the pathway is catalysed by three isoenzymes of dihydroxyacetone phosphate (DAHP) synthase, each being susceptible to one of the three end products of the aromatic pathway, phenylalanine, tyrosine or tryptophan. Control is achieved by both repression of enzyme synthesis and inhibition of enzyme activity. Within the common pathway to chorismic acid the production of shikimate kinase is also susceptible to repression. The conversion of chorismic acid to prephenic acid is catalysed by two isoenzymes of chorismate mutase, each being susceptible to feedback inhibition and repression by one of either tyrosine or phenylalanine. Each isoenzyme also carries an additional activity associated with either the phenylalanine or tyrosine branch. The tyrosine sensitive isoenzyme carries prephenate dehydrogenase activity (the next enzyme in the route to tyrosine) whilst the phenylalanine sensitive enzyme carries prephenate dehydratase (the next enzyme in the route to phenylalanine).

The regulation of gene expression was modified by:

- (i) Both tyrosine sensitive and phenylalanine sensitive DAHP synthase and shikimic kinase are regulated by the repressor protein coded by the *tyrR* gene. The *tyrR* gene had been cloned and

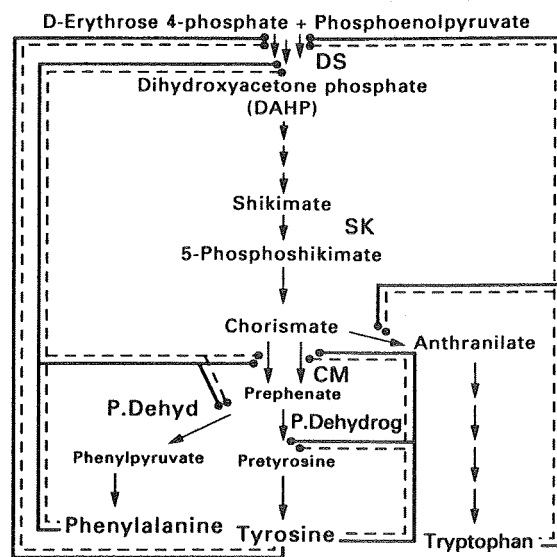


Fig. 3.30. Control of the aromatic amino acid family in *Escherichia coli*

---●	Feedback inhibition control
—●	Feedback repression control
CM	Chorismate mutase
DS	DAHP synthase
P.Dehyd	Prephenate dehydratase
P.Dehydrog	Prephenate dehydrogenase
SK	Shikimate kinase

was subjected to *in vitro* mutagenesis and the wild type gene replaced. This one mutation then resulted in the derepression of both tyrosine and phenylalanine DAHP synthase as well as shikimic kinase.

- (ii) The chorismic mutase/prephenate dehydratase protein is under both repression and attenuation control by phenylalanine. These controls were eliminated by replacing the normal promoter with one not containing the regulatory sequences, thus giving a ten times increase in gene expression.

The regulation of enzyme inhibition was modified by:

- (i) Chorismic mutase/prephenate dehydratase is subject to feedback inhibition by phenylalanine. It was shown that a certain tryptophan residue was particularly important in the manifestation of feedback inhibition. *In vitro* mutagenesis was used to delete the tryptophan codon and the modified gene introduced along with the substituted promoter referred to earlier. The enzyme

produced by the modified gene was no longer susceptible to phenylalanine.

- (ii) The rationale was to limit the availability of tyrosine such that the tyrosine-sensitive DAHP synthase isoenzyme would not be inhibited. Although the other two isoenzymes would still be susceptible to phenylalanine and tryptophan inhibition, sufficient DAHP would be synthesized by the third isoenzyme (in the absence of tyrosine) to facilitate overproduction of phenylalanine. Traditionally, this objective would have been achieved by using a tyrosine auxotroph fed with limiting tyrosine. However, these workers developed an excision vector system. An excision vector is a genetic element that can carry a cloned gene and can both integrate into, and be excised from, the bacterial chromosome. The vector is based on the bacteriophage lambda and the technology is explained in Fig. 3.31. The excision of the vector can be induced by a temperature shock. Thus, when the vector is excised the progeny of the cell will lose the inserted DNA. The *tyrA* gene was deleted from the production strain, inserted into the excision vector and transformed back into the organism where the vector became integrated into the chromosome. The fermentation could then be conducted using a cheap medium (the organism was not auxotrophic at inoculation) and allowing growth to an acceptable density. A heat shock may then be used to initiate vector excision, tyrosine auxotrophy and, hence, phenylalanine synthesis.

These efforts should have generated a high-producing strain. However, the tyrosine-sensitive DAHP synthase was susceptible to inhibition by high concentrations of phenylalanine. It will be recalled that the flow of DAHP was intended to come from the deregulated tyrosine sensitive isoenzyme. Thus, the final step in the development of the strain was to render this isoenzyme resistant to phenylalanine inhibition. This was achieved by the selection of mutants resistant to phenylalanine analogues. Thus, the strain was improved using a combination of gene cloning, *in vitro* mutagenesis and analogue resistance, indicating the importance of the contribution of a range of techniques to strain development. The final strain was capable of producing 50 g dm^{-3} phenylalanine at a yield of 0.23 g g^{-1} glucose and 2 g g^{-1} biomass. The organism produced very low

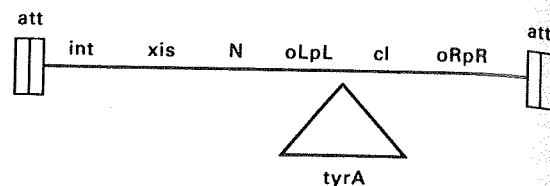


FIG. 3.31. Excision vector technology. An excision vector is represented as a line between two boxes. Above the line and boxes are indicated those genes from bacteriophage lambda that are carried on the vector. *oRpR* is required for the expression of a repressor determined by *cI*. This repressor binds to *oRpR* and *oLpL* to prevent the expression of other lambda genes. The other genes collectively act to form a recombination activity that allows the vector to integrate into or excise from the bacterial chromosome at the sites (indicated by the boxes) named *att*. Other genes, such as *tyrA* (indicated below the excision vector), can be cloned into the excision vector prior to its introduction into a target cell and can thereby be present or absent in the cell in coordination with the vector. Upon entering a cell, the genes for recombination (*N*, *xis*, and *int*) are expressed because there is not yet any repressor. As repressor accumulates, it shuts off the expression of those genes. In a fraction of the recipient cells, the recombination enzymes cause the vector to integrate into the cell chromosome before those enzymes decay away. That cell and all of its progeny inherit the vector and any gene it might carry (such as *tyrA*). If the repressor is inactivated, such as by high temperature, new recombination enzymes are formed that excise the vector from the chromosome. In such a cell and all of its progeny, the vector and the gene(s) it might carry are lost (Backman *et al.*, 1990).

amounts of the other products and intermediates of the pathway which the workers claimed was due to the very precise manipulation of the strain which avoided the concomitant adverse characteristics associated with many highly mutated organisms.

Ikeda and Katsumata (1992) redesigned a tryptophan producing *C. glutamicum* strain such that it overproduced either phenylalanine or tyrosine. The regulation of the aromatic pathway in *C. glutamicum* is shown in Fig. 3.32. Phenylalanine producers which were resistant to phenylalanine analogues were used as sources of genes coding for enzymes resistant to control. Thus, a plasmid was constructed containing genes coding for the deregulated forms of DAHP synthase, chorismate mutase and prephenate dehydratase. The vector was introduced into a tryptophan overproducer having the following features:

- (i) Chorismate mutase deficient and, therefore, auxotrophic for both tyrosine and phenylalanine.
- (ii) Wild type DAHP synthase.
- (iii) Anthranilate synthase partially desensitized to inhibition by tryptophan.

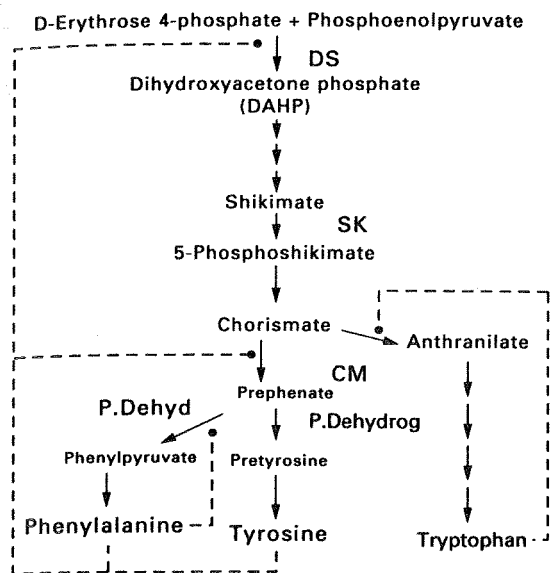


FIG. 3.32. Control of the aromatic amino acid family in *Corynebacterium glutamicum*.

- Feedback inhibition control
- CM Chorismate mutase
- DS DAHP synthase
- P.Dehyd Prephenate dehydratase
- P.Dehydrog Prephenate dehydrogenase
- SK Shikimate kinase

The transformed strain was capable of producing 28 g dm⁻³ phenylalanine and the levels of all three enzymes coded for by the vector were amplified approximately seven fold. If the original tryptophan producing strain was transformed with a plasmid containing only DAHP synthase and chorismate mutase, then tyrosine was overproduced (26 g dm⁻³). Thus, genetic engineering techniques allowed a tryptophan producer to be re-designed into either a phenylalanine or tyrosine producer.

The development of *B. sphaericus* strains overproducing biotin was discussed in our consideration of the use of analogue resistant mutants. Although *B. sphaericus* is an intrinsically good biotin producer one of the limitations to its commercial exploitation is its complex nutritional requirements (Brown *et al.*, 1991). It uses carbon sources such as glucose inefficiently and requires complex organic nitrogen sources. Thus, workers have cloned the biotin genes from *B. sphaericus* into *E. coli*. Again, it is important to appreciate that it was the analogue resistant genes that were cloned allowing deregulated production in a more commercially amenable host (Osawa *et al.*, 1989; Brown *et al.*, 1991). The two gene cassettes which were cloned were under the control of inducible promoters so that production

could be initiated after the growth phase and therefore maintain plasmid stability. Yields of up to 45 mg dm⁻³ were achieved.

A different application of recombinant DNA technology is seen in the modification of the ICI plc Pruteen organism, *Methylobacterium methylotrophus*. The efficiency of the organisms' ammonia utilization was improved by the incorporation of a plasmid containing the glutamate dehydrogenase gene from *E. coli* (Windon *et al.*, 1980). The wild type *M. methylotrophus* contained only the glutamine synthetase/glutamate synthase system which, although having a lower K_m value than glutamate dehydrogenase, consumes a mole of ATP for every mole of NH₃ incorporated. Glutamate dehydrogenase, on the other hand, has a lower affinity for ammonia but does not consume ATP. In the commercial process ammonia was in excess because methanol was the limiting substrate so the expenditure of ATP in the utilization of ammonia was wasteful of energy. The manipulated organism was capable of more efficient NH₃ metabolism, which resulted in a 5% yield improvement in carbon conversion. However, the strain was not used in the industrial process due to problems of scale-up.

The application of *in vitro* recombinant DNA technology to the improvement of secondary metabolite production may not be as advanced as it is for primary metabolites, but it has made a very significant contribution. Techniques have been developed for the genetic manipulation of streptomycetes (Hopwood *et al.*, 1985a) and the filamentous fungi (Elander, 1989) and a number of different strategies have been devised for cloning secondary metabolism genes (Hunter and Baumberg, 1989). In all the streptomycete systems so far studied the genes for the biosynthesis of a secondary metabolite are clustered. Furthermore, these clusters also contain the genes for regulation, resistance, export and extracellular processing. Work of this type has not only increased the basic understanding of the molecular genetics of secondary metabolism, but it has also facilitated strain improvement.

An excellent example of the application of recombinant DNA technology to the improvement of secondary metabolite production is provided by the work of the Lilly Research Laboratories group (Skatrud, 1992). These workers attempted to increase cephalosporin C synthesis by *Cephalosporium acremonium* by increasing the gene dosage at limiting steps in the pathway. Four critical steps were involved:

- (i) Identifying the biochemical rate limiting step in the cephalosporin C industrial fermentation.

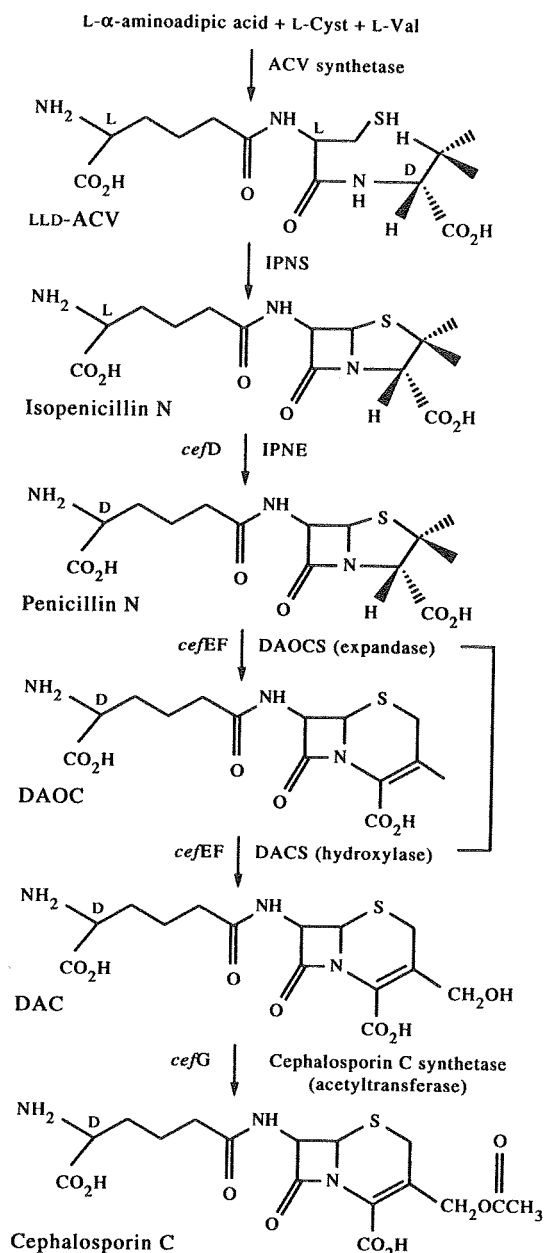


Fig. 3.33. The biosynthetic route to cephalosporin C in *Cephalosporium acremonium* (Skatrud, 1992).

Abbreviations:

- cef* Genes coding for cephalosporin synthesis
- DAC Deacetylcephalosporin C
- DACS DAC synthase (commonly known as hydroxylase)
- DAOC Deacetoxycephalosporin C
- DAOCS DAOCS synthase (commonly known as expandase)
- IPNS Isopenicillin-N-synthase

- (ii) Cloning the gene coding for the enzyme catalysing the rate limiting step.
- (iii) Constructing a vector containing the cloned gene and introducing it into the production strain.
- (iv) Screening the transformants for increased productivity.

The biosynthetic route to cephalosporin C is shown in Fig. 3.33. Analysis of fermentation broths showed an accumulation of penicillin N which indicated that there was a bottleneck at the next reaction, i.e. the ring expansion step where the 5-membered penicillin N ring is expanded to the 6-membered ring of deacetoxycephalosporin C. This step is catalysed by deacetoxycephalosporin C synthase (DAOCS, commonly called expandase) coded for by the gene *cefEF* (already previously cloned by Samson *et al.*, 1987). *cefEF* also codes for the next enzyme in the route, deacetylcephalosporin C synthase (DACS, commonly called hydroxylase). Thus, if the expandase step were rate limiting, introduction of extra copies of *cefEF* should relieve the limitation and eliminate the accumulation of penicillin N.

The production strain of *C. acremonium* was transformed with a plasmid containing an exact copy of *cefEF* and the transformants screened for increased productivity. Approximately one in four transformants were superior producers and one produced almost 50% more antibiotic in a laboratory scale fermentation. Analysis of the transformant showed that a single copy of the transforming DNA had integrated into chromosome III, whereas native *cefEF* resides in chromosome II. In pilot scale fed-batch fermentations the transformant showed a 15% increase in yield which is still a very significant increase for an industrial strain. As predicted, the transformant did not accumulate penicillin N and the bottleneck appeared to have been relieved. The superior strain was among the first eight transformants examined, whereas 10,000 survivors of a mutagen exposure rendered no improved types. This work illustrates the enormous potential of recombinant DNA technology for secondary metabolite yield improvement, but it must be appreciated that such a programme involves a very considerable financial investment and the product must be sufficiently valuable to enable a good return to be realized on that investment.

IPNE Isopenicillin-N-empimerase The brackets indicate that DAOCS and DACS enzyme activities are carried on one polypeptide coded for by the *cefEF* gene.

The major difficulty to be overcome in the use of genetically manipulated cultures is their potential instability in large-scale culture, especially continuous processes. The genetically manipulated strain of *Methylobacterium methylotrophus* appears to be stable in continuous culture, but this is not surprising since the modification renders the cell more efficient and, thus, selection force in the chemostat would tend to work to its advantage. However, strains which have been manipulated to produce products which give the cell no selective advantage would be selected against in long-term culture, especially in chemostat systems. Thus, the development of techniques to stabilize manipulated cultures and the incorporation of selectable marker genes consistent with the use of cheap fermentation media are critical to the successful fulfilment of much of the promise of genetic-manipulation techniques. The improved cephalosporin C producer described earlier was transformed with an integrating vector which means that the transformant is stable even in the absence of selective pressure.

The improvement of industrial strains by modifying properties other than the yield of product

The previous sections of this chapter have considered ways of increasing the yields of metabolites produced by industrial micro-organisms. However, the design and economics of a commercial process are influenced by properties of the organism other than its productivity. For example, although a strain may produce a very high level of a metabolite it would be unsuitable for a commercial process if its productivity were extremely unstable, or if the organism's oxygen demand were such that it could not be satisfied in the industrial fermenter available for the process. Therefore, characteristics of the producing organism which affect the process may be critical to its commercial success. Thus, it may be desirable to modify such characteristics of the producing organism which may be achieved by selecting natural and induced variants and recombinants. Naturally, it is crucial that the modified strain retains its desirable productivity so that the screening involved in these procedures should include assay of the yield as well as the characteristics being selected. Some examples of the characteristics which may be important in this context are: strain stability, resistance to phage infection, response to dissolved oxygen, tolerance of medium components, the produc-

tion of foam, the production of undesirable by-products and the morphological form of the organism.

THE SELECTION OF STABLE STRAINS

The ability of the producing strain to maintain its high productivity during both culture maintenance and a fermentation is a very important quality. Yield decay during culture storage may be avoided by the use of maintenance techniques such as those discussed earlier in this chapter, but loss of productivity during the fermentation is far more difficult to control. A decrease in the productivity of a commercial strain is normally due to the occurrence of lower-yielding, spontaneous revertant mutants which frequently have a higher growth rate than the high-producing parent, so that yield decay is especially problematical in long-term fermentations such as fed-batch or continuous culture where the faster-growing, lower producer may predominate, or even replace, the high-producing original strain. This situation is illustrated by the commercial, amino acid-producing organisms, many of which are insufficiently stable to be used in continuous-culture processes. Many workers have attempted to control the stability of amino acid-producing strains. As may be seen from the previous section on the isolation of mutants overproducing primary metabolites the amino acid producers tend to be auxotrophs, analogue-resistant mutants or revertants; such mutations removing the normal mechanisms controlling the production of amino acids. The introduction of more than one mutation giving the same phenotype may give a more stable strain since all the mutations would have to revert for the strain to lose its productivity.

Woodruff and Johnson (1970) selected a double auxotrophic mutant of *Micrococcus glutamicus* requiring both homoserine and threonine and compared its lysine-producing properties with those of a homoserine auxotroph. The authors claimed that the double mutant had a two-fold advantage in that it produced higher levels of lysine compared with the single auxotroph and was also less susceptible to reversion to low productivity, the stability of the double auxotroph being such that it was a suitable organism for the production of lysine by continuous culture.

Nakayama (1972) discussed the problem of strain reversion in the lysine fermentation and cited an example of a fermentation where 87% of the cells from a 60–88-hour culture were revertants. This very high level of revertants was probably due to the faster growth rate of the revertant compared with the single auxotroph in homoserine-limited culture. This situation was controlled, to a certain extent, by the incorporation

of antibiotics, such as erythromycin, which are more inhibitory to the rapidly growing revertants than to the homoserine limited auxotrophs. However, the use of mutants which have multiple markers appears to be a better solution, as this considerably reduces the probability of reversion to the wild type as reversion of all the markers must occur, which should be an extremely rare event. Sano and Shiio (1970) cited the use of a *C. glutamicum* mutant for the industrial production of lysine which was auxotrophic for homoserine and leucine and was resistant to *S*-(2-aminoethyl)-L-cysteine and produced 39 g dm⁻³ lysine. This strain would have had to have several reversions before it was restored to anything approaching the wild-type.

The stability of fungal diploids used in commercial fermentations may be controlled by a technique discussed by Ball (1973). Ball claimed that it may be possible to control the degeneration of a diploid into haploids by incorporating non-homologous recessive lethal mutations on separate chromosomes of an homologous pair in the diploid. The deleterious effects of the mutations would be repressed by the dominant alleles in the diploid but would be expressed in a haploid derived from the diploid, resulting in any haploids being non-viable.

A very simple but effective technique for selecting stable strains of *P. chrysogenum* was used in the Panlabs strain development programme (Lein, 1986). Final evaluation of a culture was made using the second slant of a slant-to-slant transfer. If the culture was unstable then the yield of a fermentation from the second slant would be poor, resulting in it being rejected. This procedure was followed sequentially through the programme with the result that the later strains showed less tendency to degenerate after subculturing.

Recombinant plasmids used as vectors in genetic manipulation are susceptible to two types of instability. Segregational instability is due to uneven partitioning of the plasmids at replication resulting in the production of plasmid-free daughter cells. Structural instability is a result of recombination events occurring within the vector, sometimes resulting in disruption of the desired gene. It is essential for acceptable stability that a plasmid vector contains a partitioning locus and that segregation of the plasmid is not simply a random event. The problem has been overcome in some systems by the integration of the vector into the chromosomal DNA, as already seen for filamentous fungal vectors. The disadvantage of this approach is that only one (or a few) copy of the desired gene is present whereas the use of an autonomously replicating plasmid would result in many gene copies.

THE SELECTION OF STRAINS RESISTANT TO INFECTION

Bacterial fermentations may be affected very seriously by phage infections, which may result in the lysis of the bacteria. A possible method for reducing the risk of failure due to phage contamination is to select bacterial strains which are resistant to the phages isolated in the fermentation plant (Hongo *et al.*, 1972). It is important that the apparent resistant strains isolated are not lysogenic as the carrying of a population of phages in the fermentation is a source of potential lytic phage mutants. The use of phage-resistant mutants does not ensure immunity from phage infections because new host range phages may be introduced on to the plant or phage mutants may appear. Plant hygiene is essential to minimize the risk of contamination and it is also possible to utilize chemical agents in the fermentation which selectively inhibit phage replication (Hongo *et al.*, 1972).

It may be possible to design the host organism of a recombinant fermentation such that it is more resistant to phage infection. Primrose (1990) suggested that the inclusion of one or more host-restriction and modification systems (HRM) may achieve this objective. HRM is a mechanism whereby foreign nucleic acids which enter a cell may be destroyed. Most HRM systems reduce the infectivity of bacteriophage DNA by a factor of 10² to 10⁴. The genes for many HRM systems have been cloned and they can be introduced into host strains provided, of course, that they do not degrade the foreign DNA deliberately introduced into the production strain. Also, recombinant fermentations may be made phage-resistant by changing the host organism. Although this may seem a drastic step it would be feasible if the rationale were built into the development of the process.

Infection of some fermentations with 'wild' micro-organisms may be made more easily controlled by selecting commercial strains which are resistant to various antibiotics. The antibiotics to which the commercial strain is resistant may then be used to control the level of contaminants. The danger inherent in this technique is that resistant contaminants will also tend to be selected, but reasonable protection should be given provided that stringent sterilization of a contaminated fermenter is carried out before the vessel is used again so that any antibiotic resistant contaminants are removed.

THE SELECTION OF NON-FOAMING STRAINS

Foaming during a fermentation may result in the

loss of broth, cells and product via the air outlet as well as putting the fermentation at risk from contamination (see Chapter 9). Thus, foaming is normally controlled by either chemical or mechanical means (see Chapters 4 and 7), but this task may be made easier if a non-foaming strain of the commercial organism can be developed. Foaming which occurs early in the fermentation is usually due to a component in the medium whereas foaming late in the fermentation is normally a property of the growing organism and, therefore, it is only this latter type of foam which may be controlled by strain selection. Both mutant screening and recombination may be used to develop non-foaming types but, obviously, the organisms must be tested for productivity as well. Recombination appears to be a particularly attractive proposition especially if non-foaming mutants have been isolated which are poor producers or good producers which foam. Ancestral crosses using protoplast fusion techniques may then render a strain combining the desirable features of both strains.

THE SELECTION OF STRAINS WHICH ARE RESISTANT TO COMPONENTS IN THE MEDIUM

Some media components which are required for product formation may interfere with the growth of the organism and, therefore, it may be desirable to select for strains which are resistant to the medium component. Polya and Nyiri (1966) applied this approach to the isolation of mutants of *P. chrysogenum* resistant to phenylacetic acid, a precursor of penicillin and toxic to the organism at high concentrations. Barrios-Gonzalez *et al.* (1993) used a very similar approach in isolating superior penicillin producers. Alikhanian *et al.* (1959) selected strains of the oxytetracycline producer, *S. rimosus*, resistant to high levels of phosphate which prevents product synthesis. This enabled the strain to be used in media containing high levels of corn steep liquor (which is rich in phosphate).

Analogues of repressing media components have been used to select resistant mutants. For example, arsenate and vanadate have been used to isolate phosphate resistant strains (DeWitt *et al.*, 1989) and deoxyglucose to isolate glucose catabolite repression resistant strains.

THE SELECTION OF MORPHOLOGICALLY FAVOURABLE STRAINS

The morphology of a micro-organism in submerged culture frequently has an effect on the economics or ease of operation of a process. The morphological form

of a filamentous micro-organism will affect both the aeration of the system and the ease of filtration of the fermentation broth. As discussed in Chapter 6, the morphology of a fungus in submerged culture may be controlled by the level of a spore inoculum and by the medium, but it is also possible to influence the organism's morphology by altering its genotype. Backus and Stauffer (1955) recognized the influence of the genetics of a strain on the morphology of *P. chrysogenum* in submerged culture and its role in controlling foaming and broth filtration characteristics.

Bartholomew *et al.* (1977) selected strains of *P. chrysogenum* (for penicillin production) which gave a lower viscosity broth which increased the oxygen-transport ability of their fermentation plant equipment. The use of mutation and screening compared with recombination for the production of strains of a particular morphological type has been discussed by Ball (1978). Ball claimed that recombination techniques would be at an advantage over mutation and selection because of the large number of mutants that would have to be screened. A recombination programme would involve the crossing of a strain of the required morphology with a commercial producer, which should involve the screening of relatively few progeny. Hamlyn and Ball (1979) applied the technique of protoplast fusion to the construction of a desirable strain of *C. acremonium* for the production of cephalosporin, as discussed in an earlier section. A cross between an asporulating, slow-growing strain with a sporulating, fast-growing strain which only produced one third of the cephalosporin level of the first strain eventually resulted in the isolation of a recombinant which displayed the desirable properties of both strains, i.e. a strain which demonstrated good sporulation, a high growth rate and produced 40% more antibiotic than the higher-yielding parent. The ability of a strain to sporulate profusely is a very useful characteristic in the development of inoculum, as discussed in Chapter 6.

Rowlands (1992) reported that genetically altering *P. chrysogenum* strains to produce a pelleted rather than a filamentous broth was an important feature in the Panlabs penicillin improvement programme. The pelleted form gives rise to a much lower broth viscosity, resulting in lower power consumption. Although Rowlands did not indicate which genetic technique was used (mutant selection or protoplast fusion), protoplast fusion was certainly used to improve the sporulation and growth of the Panlabs strain, as discussed in a previous section (Table 3.7).

Flocculation of yeasts is the adherence of cells in

clumps resulting in the separation of the cells from the liquid in which they were suspended. Thus, the flocculating property of yeasts may be described as a morphological characteristic. When flocculation occurs in a beer fermentation the yeast will rise to the top of the vessel if it is a 'top-fermenting' yeast or drop to the bottom of the vessel if it is a 'bottom-fermenting' yeast. Thus, the flocculence of a yeast will determine the time of contact of the yeast with the wort (and hence the conversion of the wort to alcohol) and the ease of clarification of the beer. Although mutant selection is rarely applied, the selection of natural variants is a very common practice in many breweries. As discussed in Chapter 6, the yeast produced from one beer fermentation may be used to inoculate a new fermentation, a practice which is very rarely employed in the rest of the fermentation industry. By the careful selection of the yeast produced during a fermentation, strains showing the desired degree of flocculence may be selected. The first cells to rise to the top of a top fermentation, or drop to the bottom of a bottom fermentation, are the most flocculent cells whereas the last to rise or drop (depending on the type of fermentation) are the least flocculent cells. Thus, by selecting those cells that flocculate at an intermediate time the brewer isolates the natural variants which have the most desirable morphological properties for the inoculation of the next fermentation.

THE SELECTION OF STRAINS WHICH ARE TOLERANT OF LOW OXYGEN TENSION

The provision of oxygen is frequently the limiting factor of many fermentations and it would, therefore, be desirable to select an organism which was capable of producing the product at a lower oxygen tension than normal. This may be achieved by screening for increased production under oxygen limited conditions. For example, Mindlin and Zaitseva (1966) isolated a lysine-producing strain which maintained its productivity under aeration conditions which decreased the parental strain productivity by almost a half.

THE ELIMINATION OF UNDESIRABLE PRODUCTS FROM A PRODUCTION STRAIN

Although an industrial micro-organism may produce large quantities of a desirable metabolite it may also produce a large amount of a metabolite which is not required, is toxic or may interfere with the extraction process. Thus, in these circumstances it would be an advantage to alter the strain such that the undesirable product is no longer produced. An example in the

penicillin-producing strains is the elimination of the production of the yellow pigment, chrysogenein, by the selection of non-pigmented mutants which made the extraction of the antibiotic much simpler (Backus and Stauffer, 1955).

Dolezilova *et al.* (1965) considered the production of fungicidin (nystatin) and cycloheximide by mutants of *S. noursei*. These workers demonstrated that mutants could be isolated which produced increased levels of fungicidin but produced no cycloheximide. The use of protoplast fusion techniques for the elimination of *p*-OH penicillin V has been considered earlier in the chapter.

THE DEVELOPMENT OF STRAINS PRODUCING NEW FERMENTATION PRODUCTS

The isolation of organisms from the natural environment synthesizing commercially useful metabolites is an expensive and laborious process. Therefore, other means of producing novel compounds which may be of some industrial significance have been attempted. Probably the most successful alternative approach has been the semi-synthetic one where microbial products have been chemically modified, for example, the semi-synthetic penicillins. Precursor feeding has also met with some success in this context; by incorporating a precursor of a natural product into the fermentation medium the level of the end product may be increased, for example, the use of phenylacetic acid in the production of penicillin G. The feeding of an analogue of the normal precursor of a natural product frequently results in the production of an analogue of the natural product. Hamill *et al.* (1970) demonstrated that if 5-, 6- or 7-substituted tryptophan replaced tryptophan, the normal precursor of the antifungal agent, pyrrolnitrin, in cultures of *Pseudomonas aureofaciens*, then a series of substituted pyrrolnitrins were obtained, some of which had improved antifungal activity. However, the disadvantages of the analogue-precursor technique are that the end product tends to be very similar to the natural product and the new product will be contaminated with the normal product which the organism may still synthesize from its self-produced natural precursors. Birch (1963) suggested that the problem of mixed products may be overcome by isolating mutants which would not produce the normal precursor but could convert it into the end product. Thus, the analogue precursor could be converted into the novel end product without competition from the normal endogenous precursor. Shier *et al.* (1973) succeeded in applying Birch's idea in the study of neomycin production by *Strept-*

tomyces fradiae from the precursor, deoxystreptamine. By using a replica plating technique, the survivors of a mutation treatment were screened for the ability to inhibit the growth of the test organism, *Bacillus subtilis*, only in the presence of deoxystreptamine. By feeding the mutant isolated by this method, different antibiotics were synthesized.

Nagaoka and Demain (1975) described this technique for the isolation of new products as 'mutational biosynthesis' and the mutants isolated as 'idiotrophs'. Besides the aminoglycoside-aminocyclitol antibiotics (of which neomycin is an example), mutational biosynthesis has been applied to the macrolide antibiotics, the novobiocin antibiotics and the β -lactam antibiotics (Daum and Lemk, 1979).

The advent of readily available recombination techniques resulted in attempts to produce novel compounds from recombinants, particularly streptomycetes. The rationale behind these experiments was that by mixing the genotypes of two organisms synthesizing different metabolites then new combinations of biosynthetic genes, and hence pathways, may be produced. Little progress was achieved using protoplast fusion but the exploitation of recombinant DNA technology has yielded some significant successes.

Streptomyces coelicolor produces the polyketide actinorhodin (Fig. 3.34) whilst *Streptomyces* sp. AM 7161 produces medermycin. Hopwood *et al.* (1985b) transformed some of the cloned genes coding for actinorhodin into *Streptomyces* sp. AM7161. The recombinant produced another antibiotic, mederrhodin A (Fig. 3.34). The modified strain contained the *actV* gene from *S. coelicolor* coding for the *p*-hydroxylation of actinorhodin; the enzyme was also capable of hydroxylating medermycin. Hopwood *et al.* (1985b) also introduced the entire actinorhodin gene cluster into *Streptomyces violaceoruber* which produces granaticin or dihydrogranaticin (Fig. 3.35). The recombinant synthesized the novel antibiotic, dihydrogranatirhodin which has the same structure as dihydrogranaticin apart from the stereochemistry at one of its chiral centres (Fig. 3.35).

The enzymes responsible for polyketide biosynthesis have been extensively studied in recent years and significant advances have been made in the understanding of both their biochemistry and genetics. Polyketide synthases (PKSs) are multifunctional enzymes which catalyse repeated decarboxylative condensations between coenzyme A thioesters and are very similar to the fatty acid synthases. An enormous range of microbial polyketides are known and the variation is due to the chain length, the nature of the precursors and

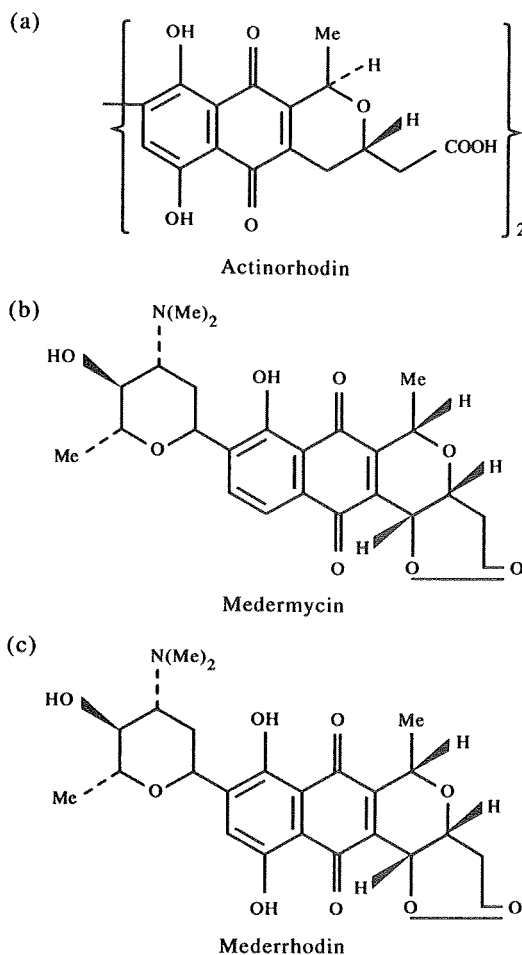


FIG. 3.34. (a) Structure of actinorhodin, (b) structure of medermycin, (c) structure of mederrhodin (Hunter and Baumberg, 1989).

the subsequent modification reactions which occur after cyclization. McDaniel *et al.* (1993) have constructed plasmids coding for recombinant PKSs and achieved expression in *Streptomyces coelicolor*. Five novel polyketides were synthesized and the system shows great promise for the production of new, potentially valuable, compounds as well as providing carbon skeletons amenable to derivatization by organic chemists.

A commercially relevant example is provided by the Lilly Research Laboratories group (Beckman *et al.*, 1993) who produced a strain of *Penicillium chrysogenum* capable of accumulating deacetoxycephalosporin C (DAOC), a compound which can be biotransformed into 7-aminodesacetoxycephalosporanic acid